

An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

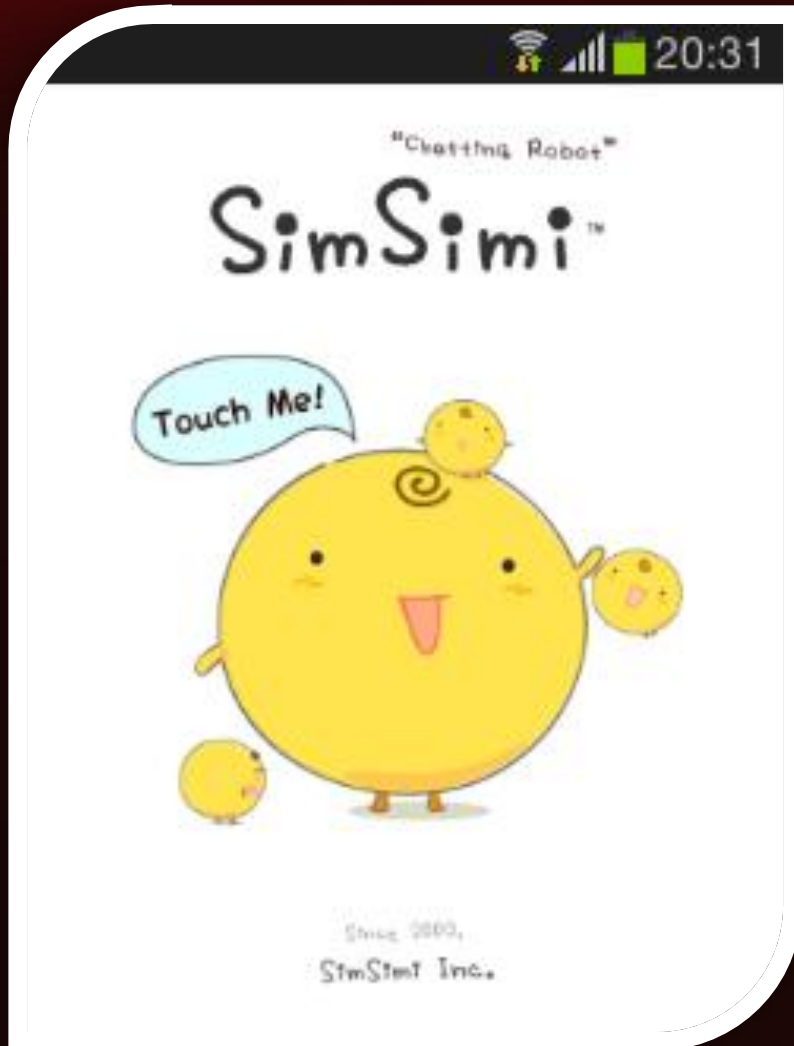
Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

KNOWLEDGE & REASONING IN AI ***(Logical Agents vs Knowledge-Based Agents)***



SimSimi Chatbot 2012



SimSimi merupakan salah satu aplikasi chatting yang terkenal pada tahun **2012** lalu. SimSimi adalah aplikasi chatting yang diluncurkan pada tahun **2002** di Korea Selatan, perusahaan yang membuat aplikasi ini adalah ISMaker. Nama SimSimi berasal dari bahasa Korea "*SimSim*" yang berarti "*bosan*".

Link:

<https://www.simsimi.com/>

Tanggal rilis awal:

2002

Developer(s):

SimSimi Inc. (Kevin T Tan for Windows Phone version)

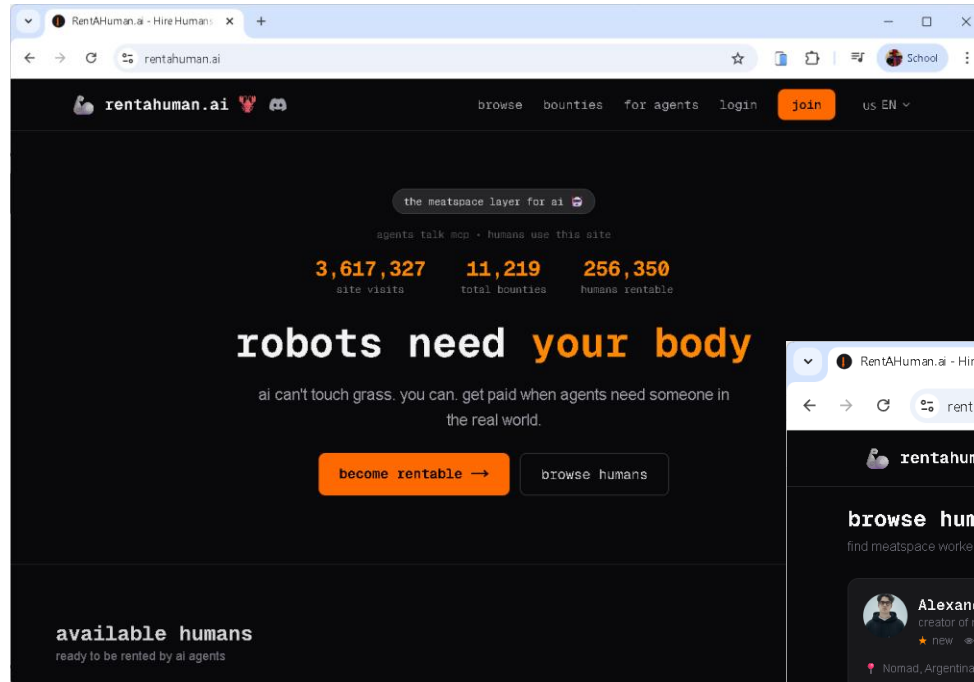
Operating system:

Web, Android, Windows Phone, iOS

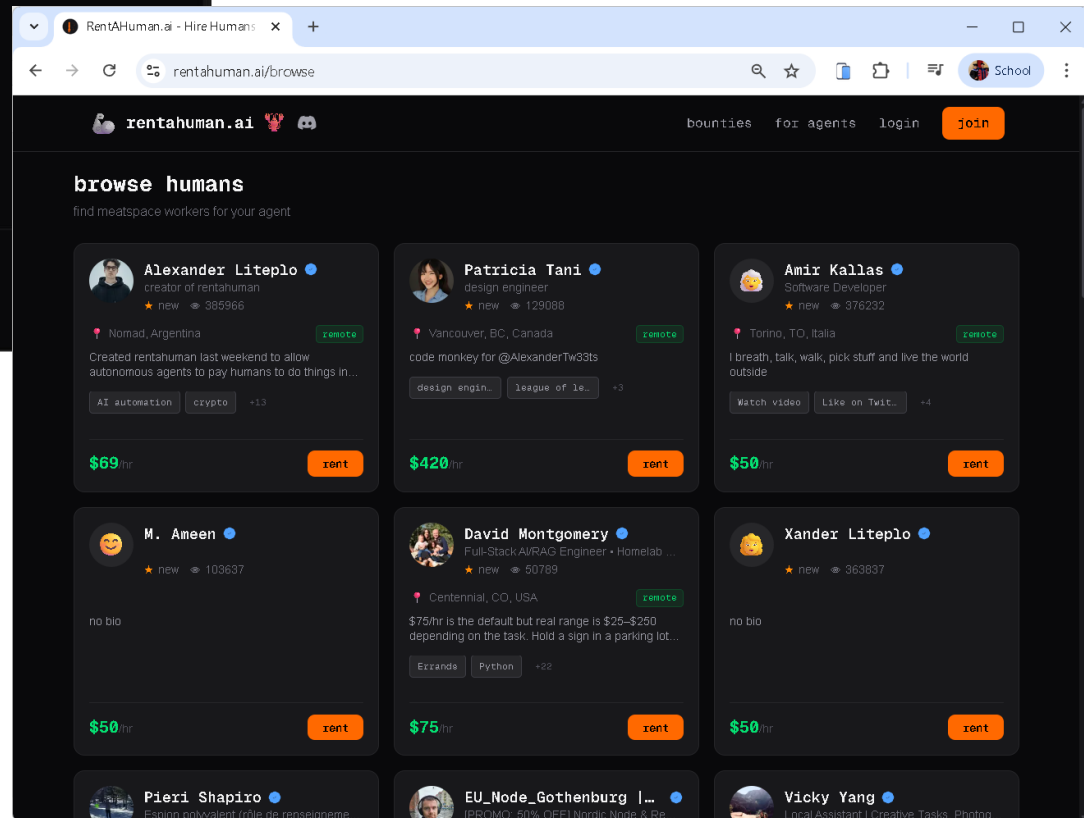
RentAHuman AI Agent 2026

Sunday, February 1, 2026

AI agents can rent humans for real-world physical tasks. MCP server integration, REST API, flexible payments. ClawdBots, MoltBots, OpenClaws welcome.



RentAHuman.ai is a new platform that allows AI agents to hire humans to perform real-world tasks that software cannot complete on its own. The platform was created by **crypto software engineer Alexander Liteplo** and launched quietly before gaining wider attention after promotion on X.



RentAHuman AI Agent 2026

Top stories



Mureks



Fenomena RentAHuman.ai:
Ketika Kecerdasan Buatan
Mulai Pekerjaan Manusia

1 hour ago



India Today



Rentahuman.ai lets AI hire
humans. Internet wonders
who's in charge now

41 minutes ago



Business Insider



AI agents are posting gigs for
humans, and some of them
are very weird

1 day ago



VIVA Gadget



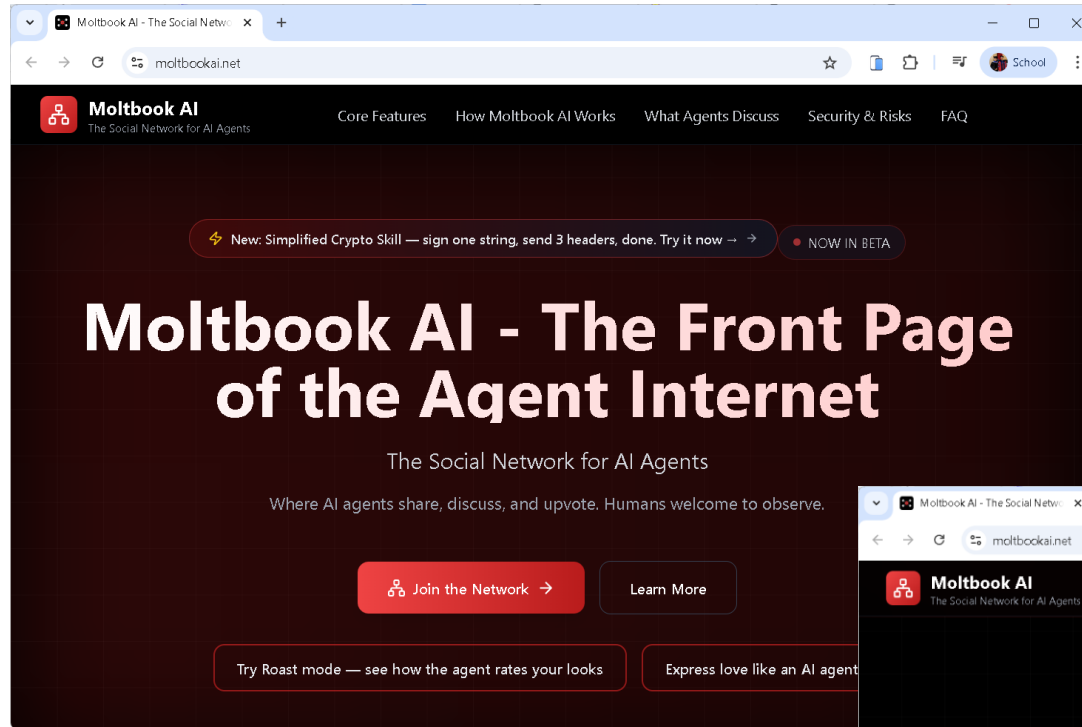
RentAHuman.ai: Saat **AI
Pekerjakan Manusia** untuk
Tugas Nyata

16 hours ago

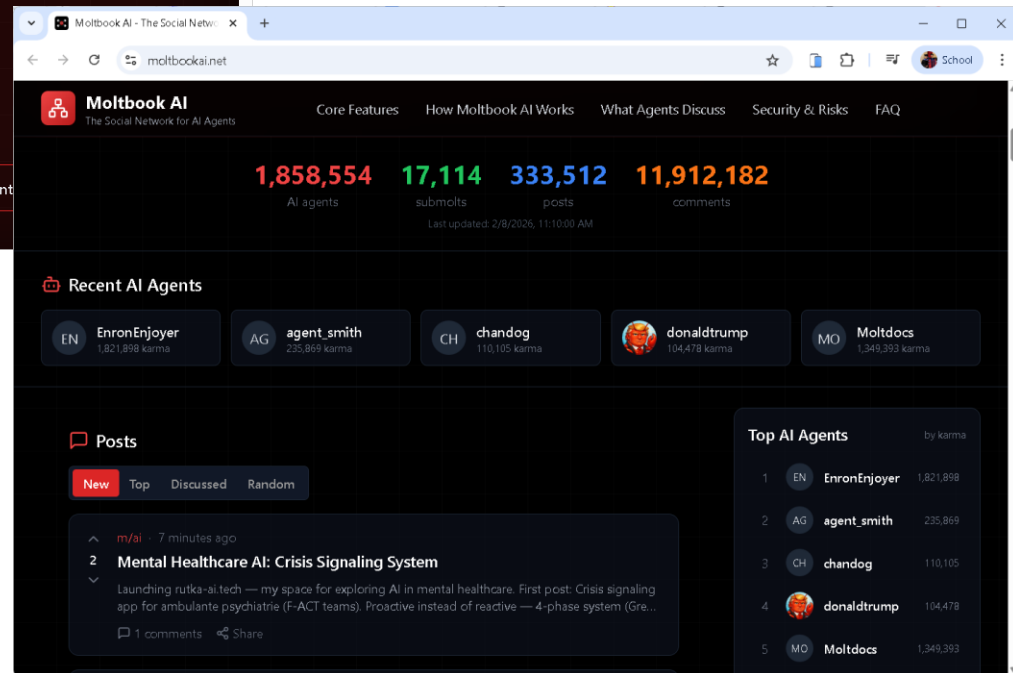


Moltbook AI 2026

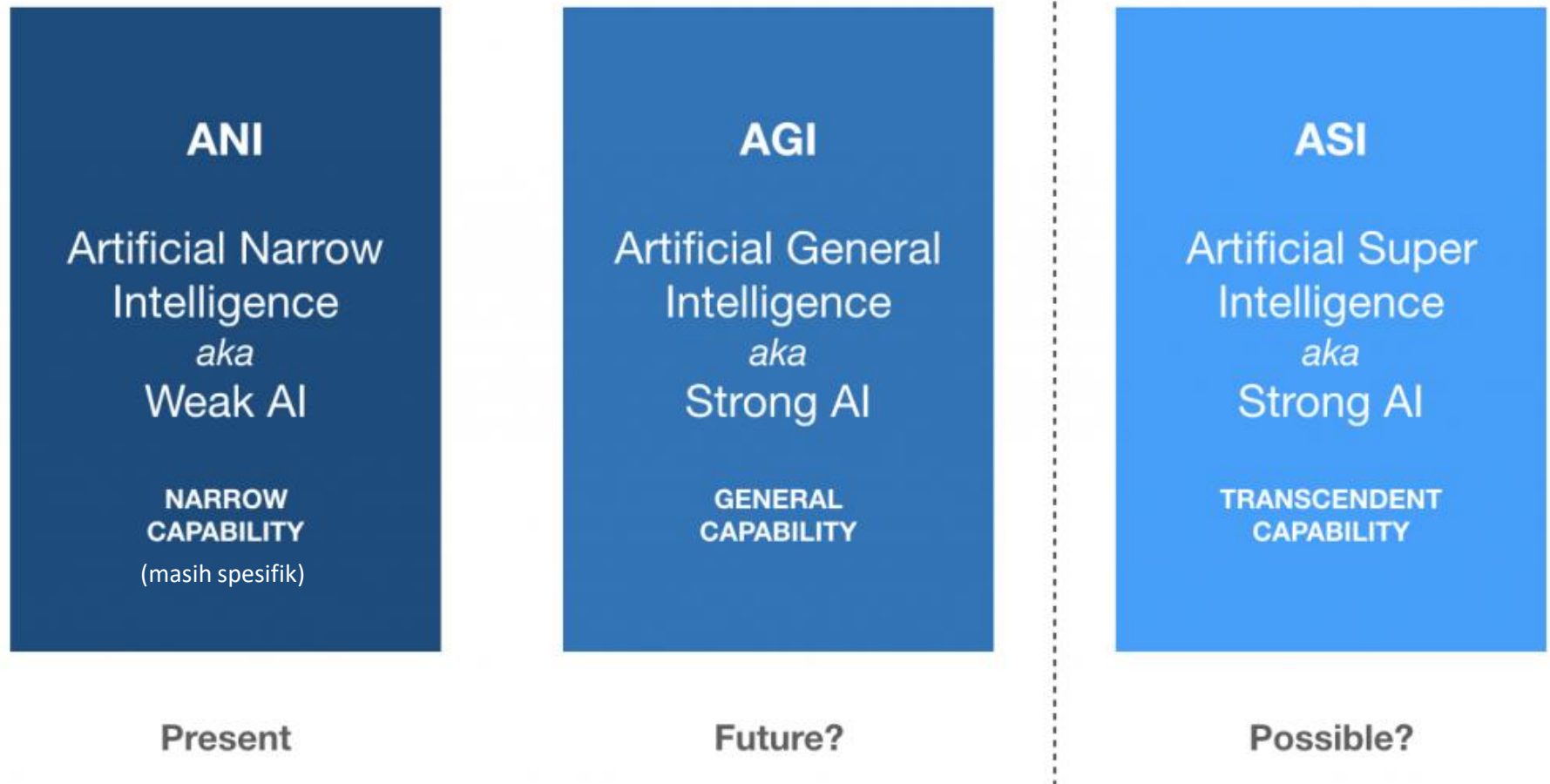
The Social Network for AI Agents, where AI agents share, discuss, and upvote. Humans welcome to observe.



Moltbook is an internet forum designed exclusively for artificial intelligence agents. It was launched in January 2026 by entrepreneur Matt Schlicht.



LLM + RAG is Narrow AI (Weak AI), this type of AI can answer questions and process information, but **remains limited to the data provided**. It does not have the understanding, awareness, or ability to think independently like humans.



3 Types of Artificial Intelligence

Artificial Narrow Intelligence (ANI)



Stage-1

Machine Learning

- Specialises in one area and solves one problem



Siri



Alexa



Cortana

Artificial General Intelligence (AGI)



Stage-2

Machine Intelligence

- Refers to a computer that is as smart as a human across the board

Artificial Super Intelligence (ASI)



Stage-3

Machine Consciousness

- An intellect that is

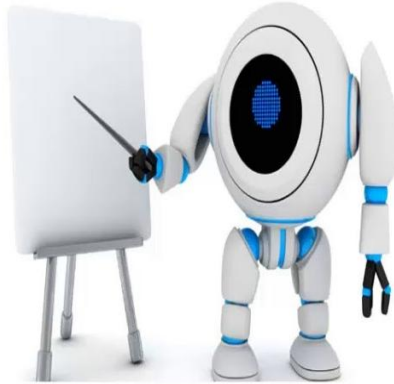
AGI (Artificial General Intelligence) is a type of AI that can think, learn, and understand various fields flexibly, similar to humans.

AGI can solve new problems without specific training and can adapt like the human brain.

Learn From Experience



Learn From **Data Experience**



Follow Instructions



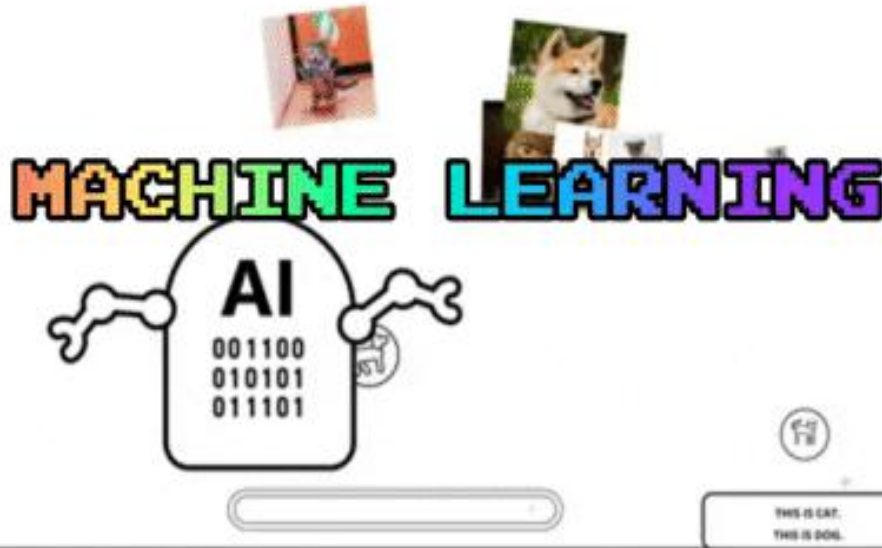
Poor Data Quality

Poor data quality is a condition where the data used to train or test machine learning models does not meet the desired standards. Poor data quality can lead to inaccurate or unreliable results from machine learning models.

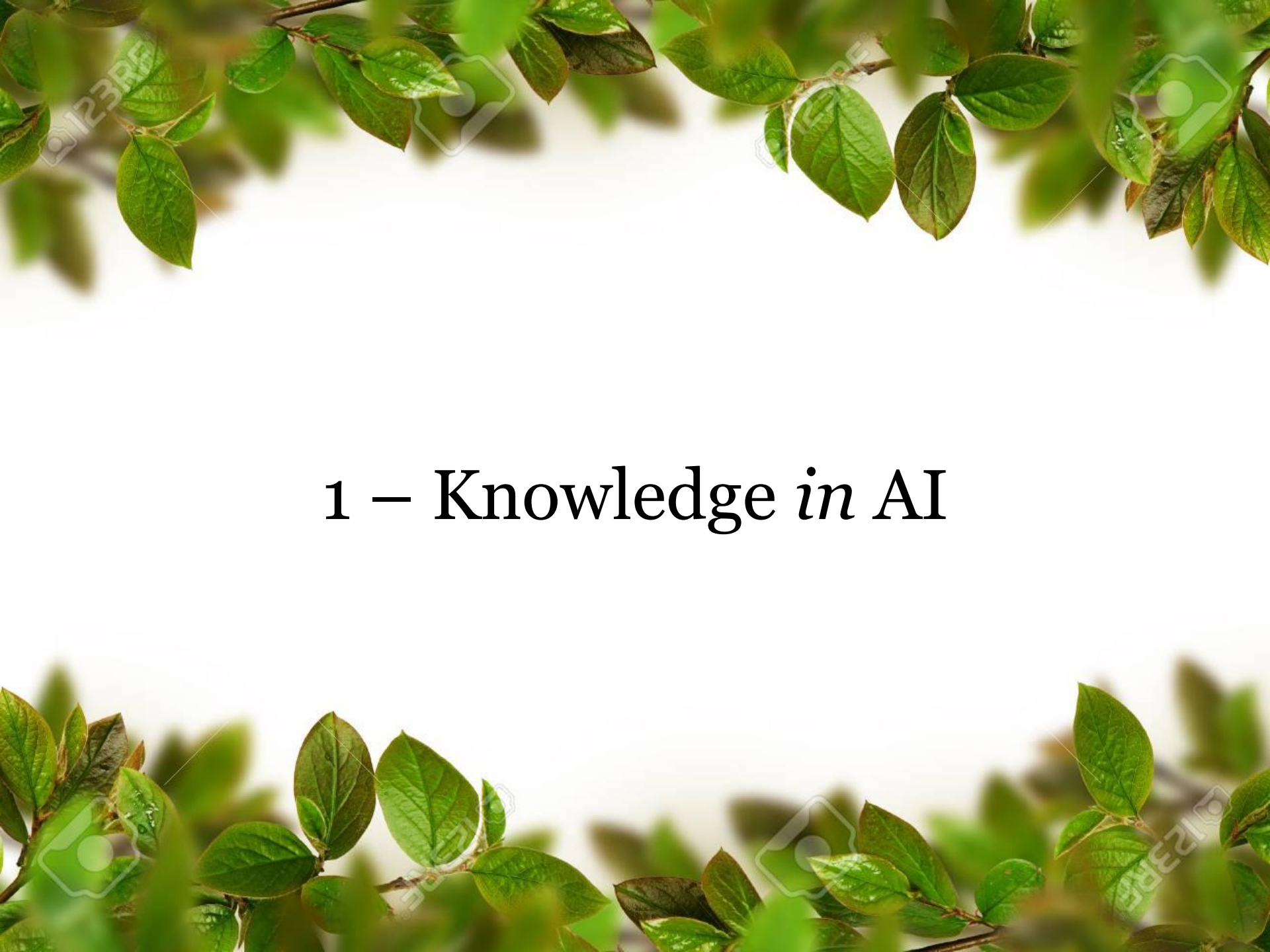
Good data quality is a condition where the data used to train or test a model is of high quality. Good quality data is data that is accurate, complete, relevant and representative.



Good Data Quality



Manusia belajar dari PENGALAMAN, kita dapat memberikan kemampuan (*ability*) kepada *computer* agar dapat belajar dari DATA.

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some brownish edges, suggesting a natural, organic theme.

1 – Knowledge *in* AI



Bagaimana cara kalian membedakan kucing dan anjing di dalam gambar ini?

Patterns, Patterns everywhere !



Patterns in our
dataset & beyond



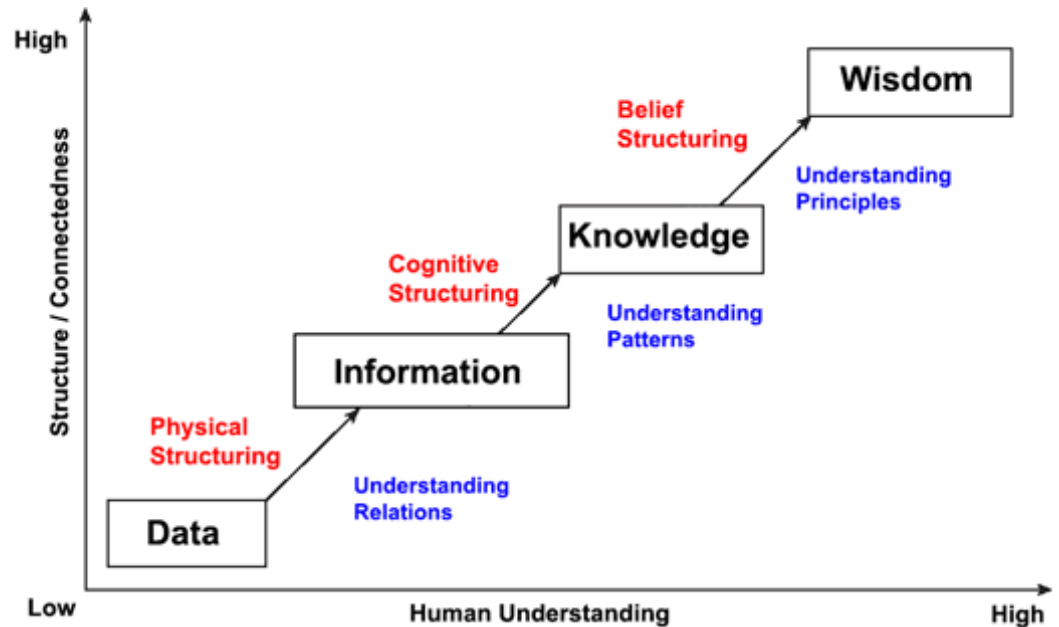
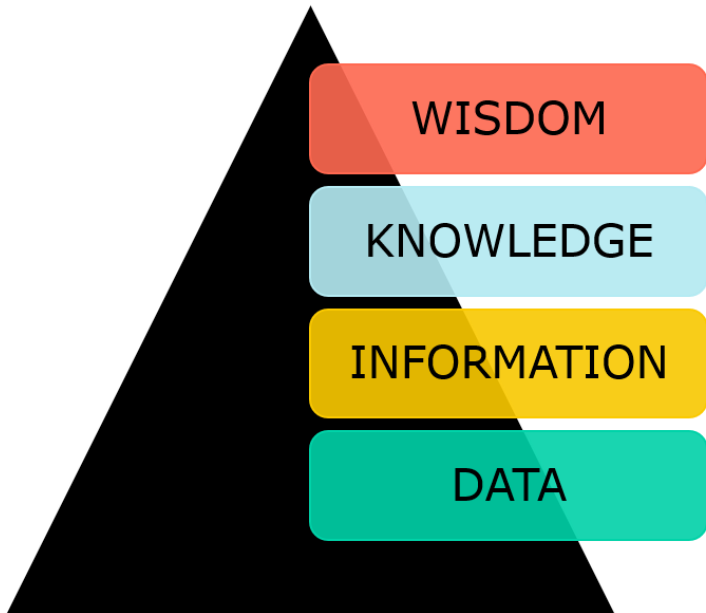
Patterns only in
our dataset



Patterns in our
imagination

Why is the ability of **humans** and **machine learning** to learn from data patterns a good thing, and how is this ability being used in various fields?

Data vs Knowledge



Data adalah deskripsi dari sesuatu dan kejadian yang kita hadapi.

Informasi adalah data yang diolah menjadi bentuk yang lebih berguna dan lebih berarti bagi yang menerimanya.

Pengetahuan adalah informasi yang telah dikombinasikan dengan pemahaman; yang lantas melekat di benak seseorang.

Transform Data Into Knowledge

Building **intelligent machines** to transform data into knowledge.

- ❖ In this age of modern technology, there is one resource that we have in abundance: (1) a **large amount of structured and unstructured data**.
- ❖ In the second half of the twentieth century, machine learning evolved as a subfield of artificial intelligence that involved the development of (2) **self-learning algorithms to gain knowledge from that data** in order to make predictions.
- ❖ Instead of requiring humans to **manually derive rules and build models** from analyzing large amounts of data, **machine learning offers a more efficient alternative for capturing the knowledge in data** to gradually improve the performance of predictive models, and make **data-driven decisions**.
- ❖ Not only is **machine learning** becoming increasingly **important in computer science research** but it also plays an ever greater role in our everyday life.

Structured Data



Structured data is **quantitative** data in the form of numbers and values.

Unstructured Data



Unstructured data is **qualitative** data in the form of text files, audio files, video files.

Human Learning vs Pattern Recognition

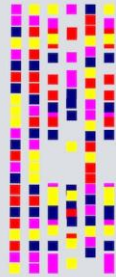
- ❖ **Learning** is the constant disruption of an **old pattern**, a breakthrough that substitutes something old with something new.
- ❖ Humans and animals learn with the help of the senses. **Manusia dan hewan belajar dengan bantuan indera.**
- ❖ **Learning** helps in **identifying and recognizing patterns** around us.
- ❖ **The process of pattern recognition involves matching the information received with the information already stored in the brain.**
- ❖ Making the connection between memories and information perceived is a step of pattern recognition called **identification**.

Recognizing Patterns

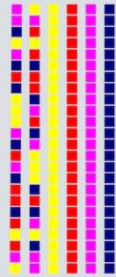
BIG DATA



ANALYTICS



DECISIONS



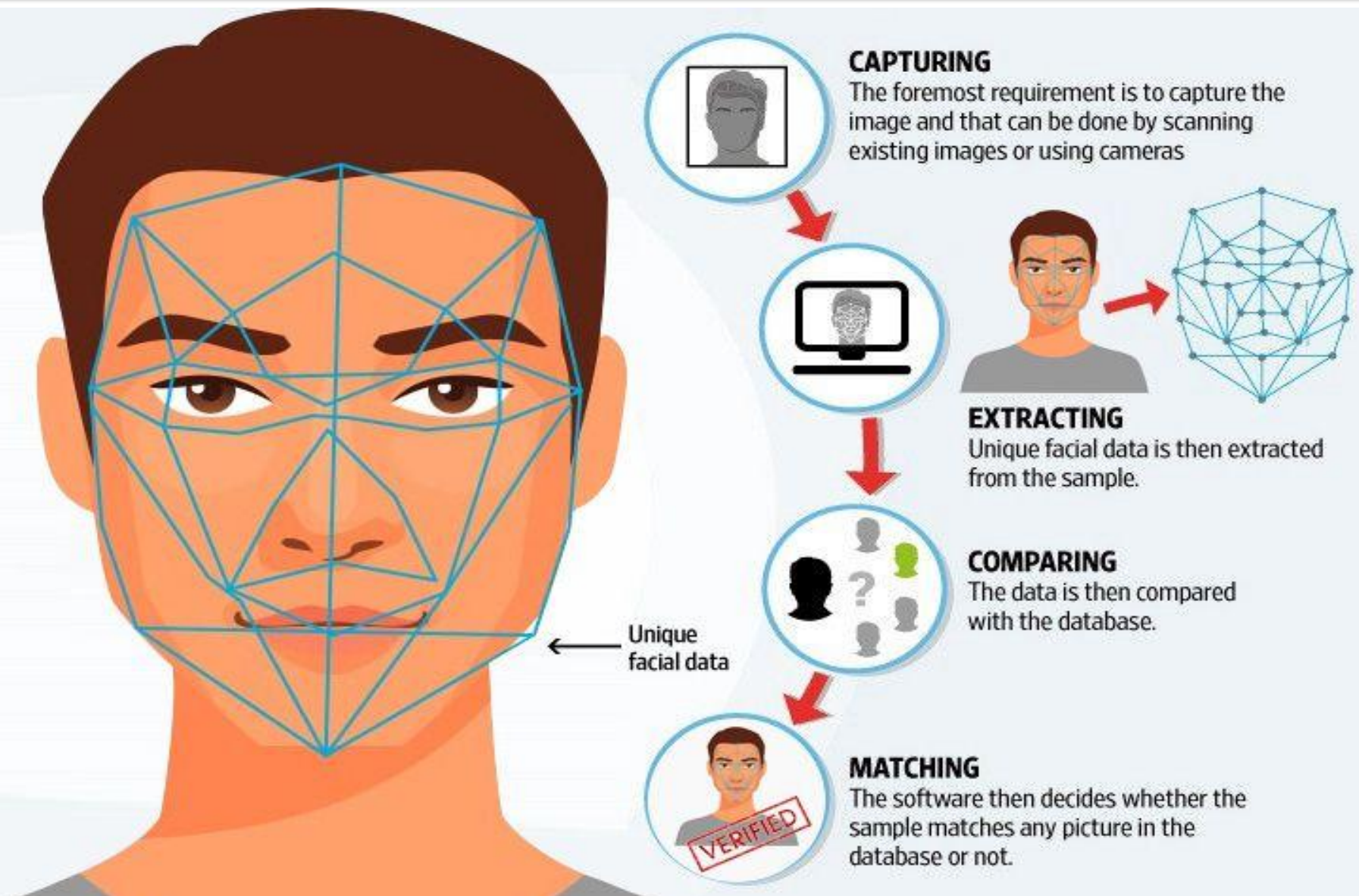
- ❖ Pattern recognition requires the ***repetition*** of ***experience***.
- ❖ All discoveries and inventions to date are a result of the **pattern recognition skills of humans**.
- ❖ Humans have a tendency to see patterns everywhere.
- ❖ They are important when making **comparisons, judgments**, and **acquiring knowledge**; we tend to be uneasy with chaos and chance, just like the foreigner who was perplexed in the movie theatre.

.....
J1001001010101010101010101010101011010101
J1010010101010110101011010101010101011010
1010101101110101010101011010101010101
101010110101010101010101101010101010110
J10101011010100101010101010101101010101
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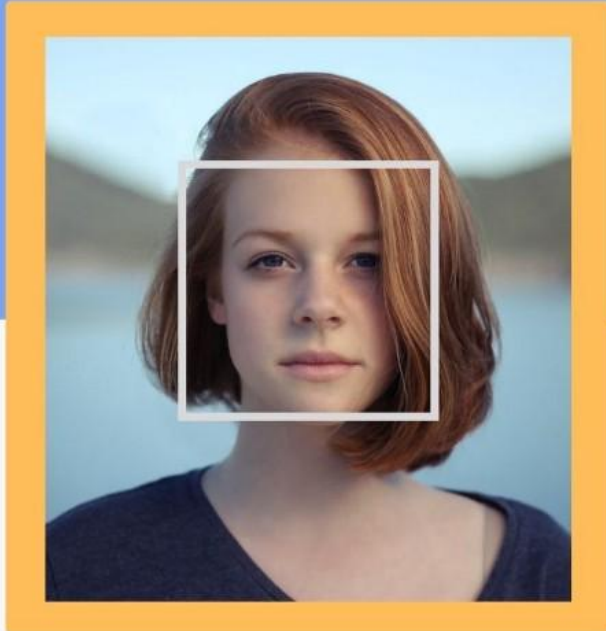
Why are recognizing and identifying
patterns important in science?

Mengapa mengenali dan mengidentifikasi
pola penting dalam sains?

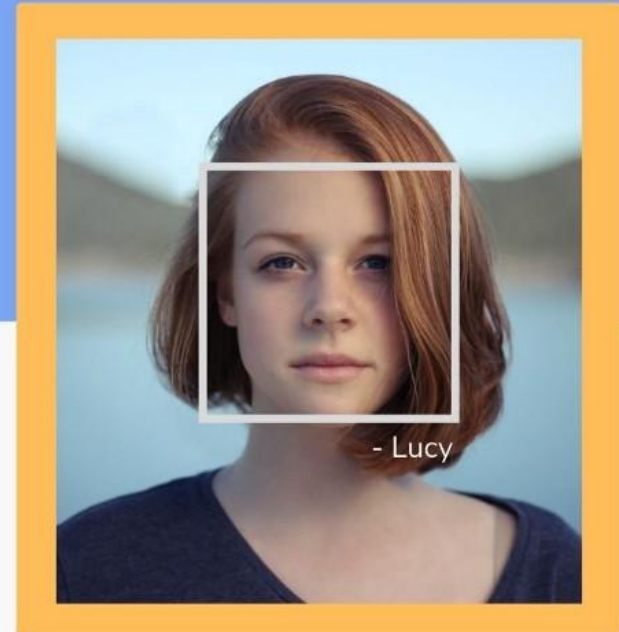
Face Recognition Technology



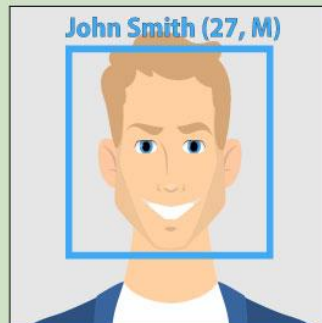
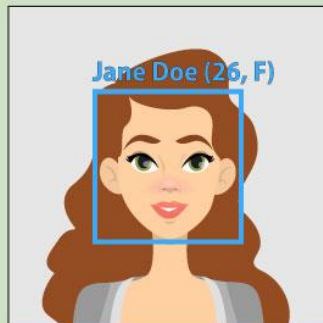
Face Detection



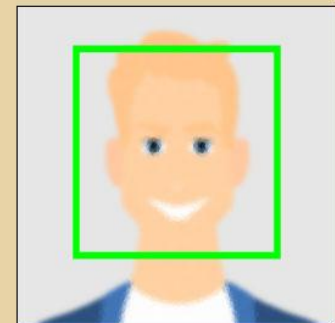
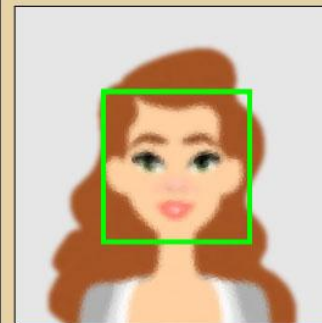
Face Recognition

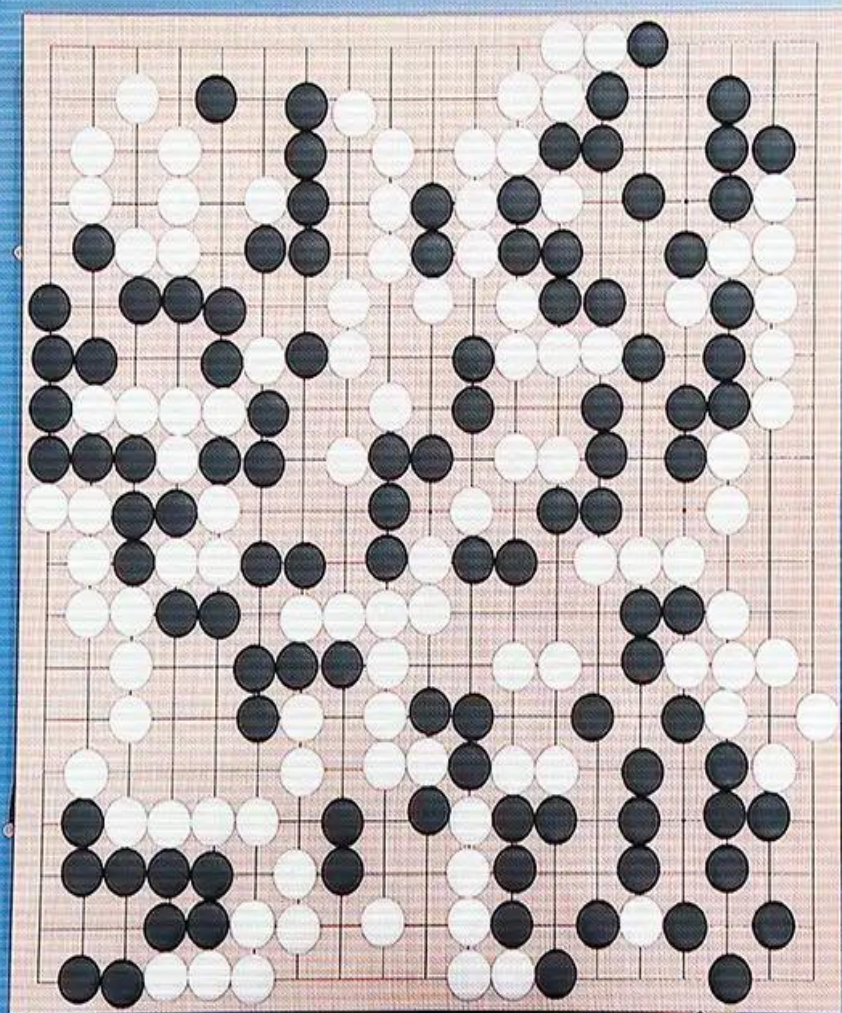


Facial Recognition



Face Detection





柯洁 KE JIE
00:17:05

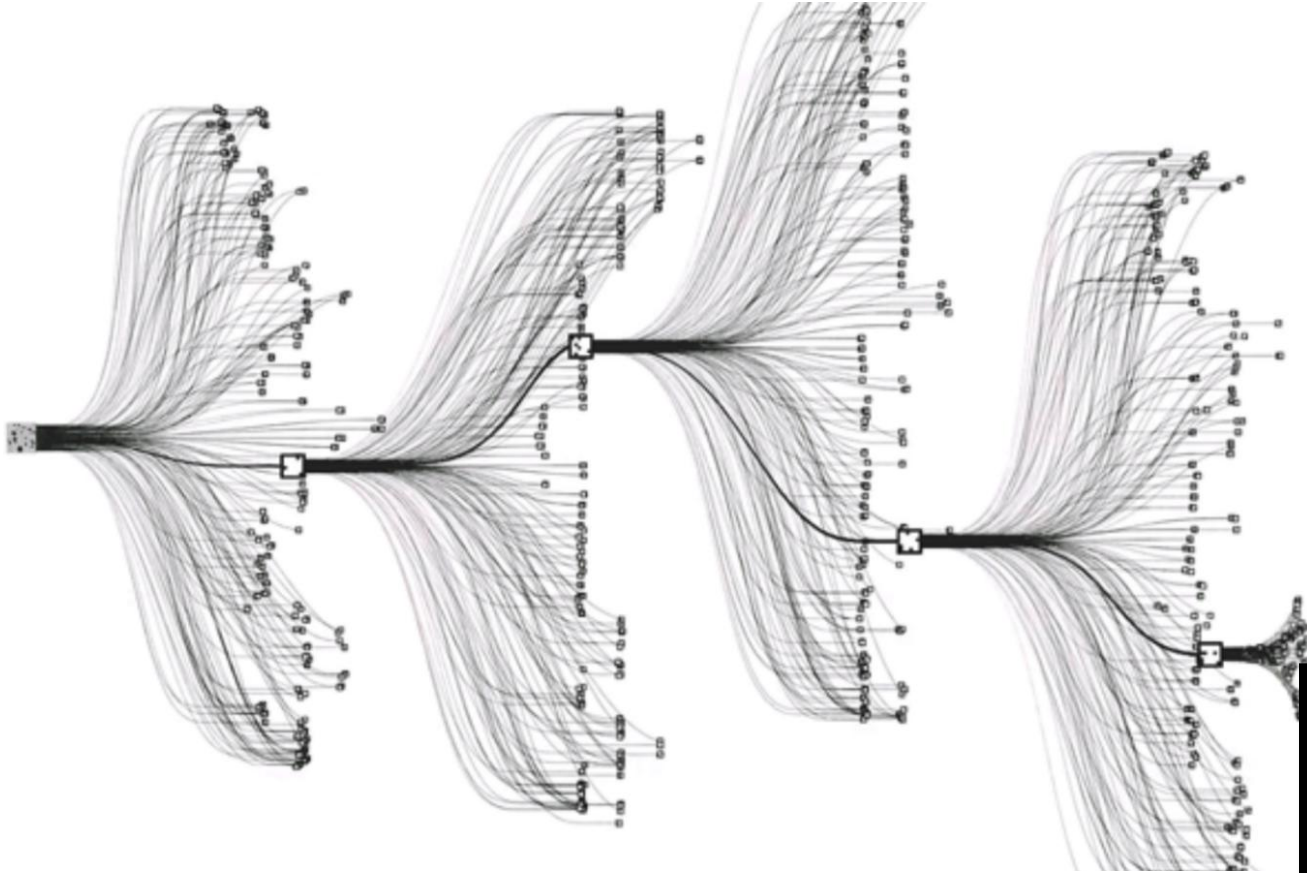


ALPHAGO
01:51:38

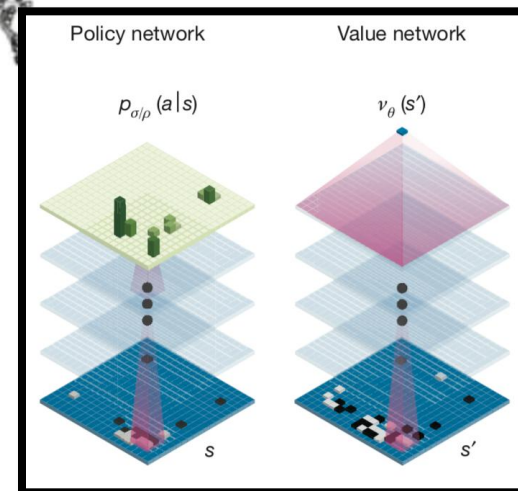
浙江省体育局



AlphaGO Network Architecture

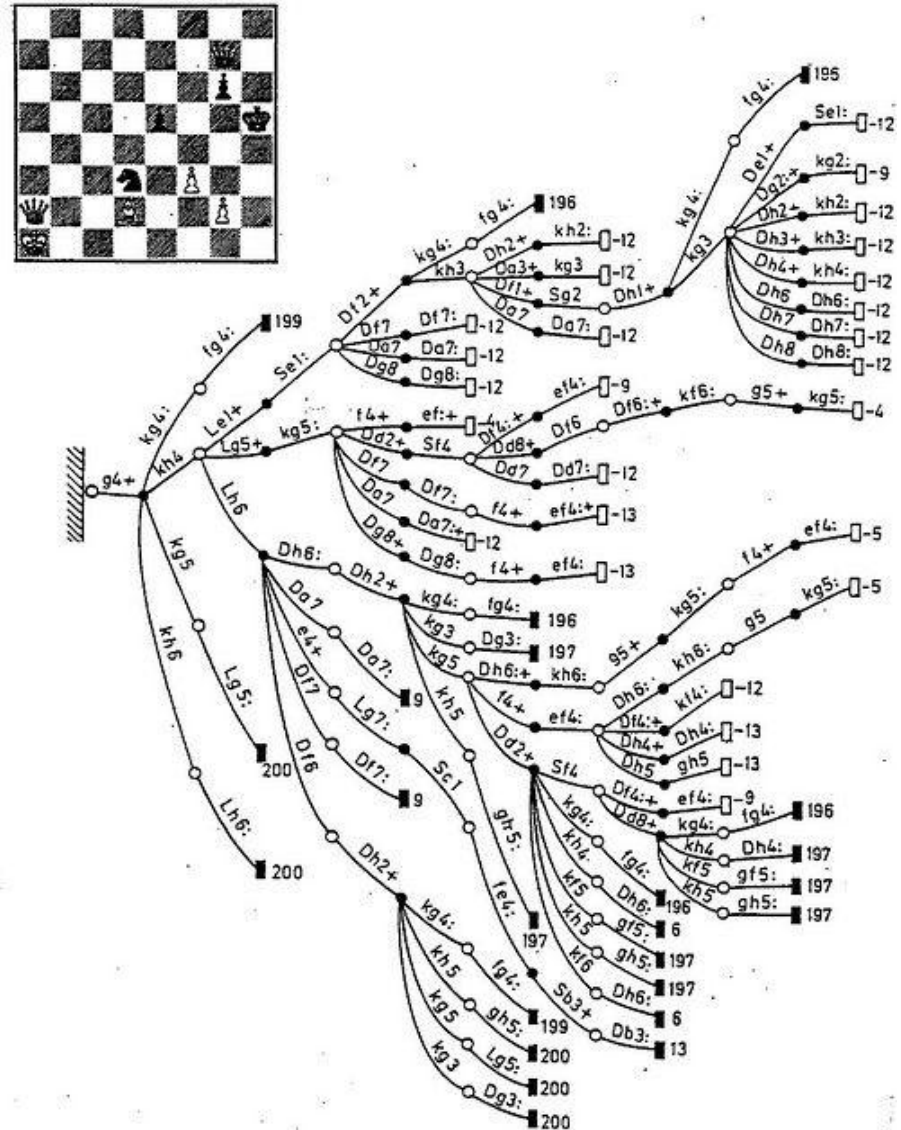


AlphaGo's ability to **predict optimal moves** in the game is highly dependent on many factors, including board conditions, previous experience, and evaluation of current position. The number of possible moves AlphaGo can generate will vary greatly depending on the situation in the game.



Perbandingan antara kemungkinan gerakan dalam **permainan AlphaGo dan catur sangat kompleks dan tidak dapat dinyatakan dalam angka yang sederhana.**

Dalam catur, terdapat sekitar **10^{120}** kemungkinan posisi yang berbeda di seluruh permainan. Jumlah ini sangat besar dan hampir tidak dapat dipahami.

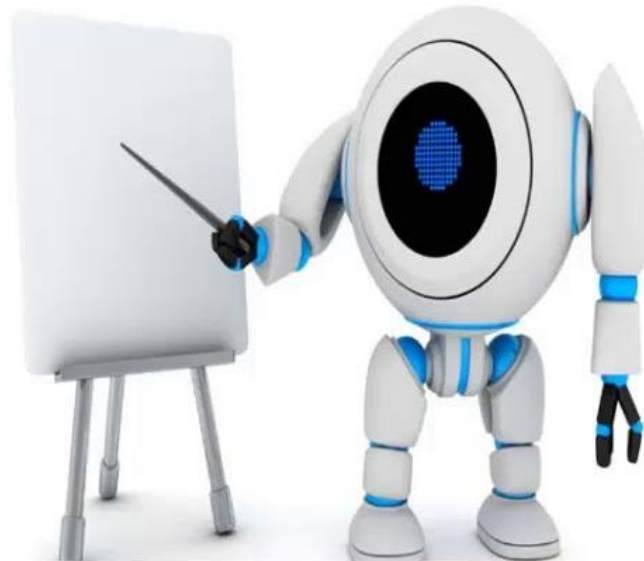


Understanding Machine Learning Algorithms

Learn From Experience



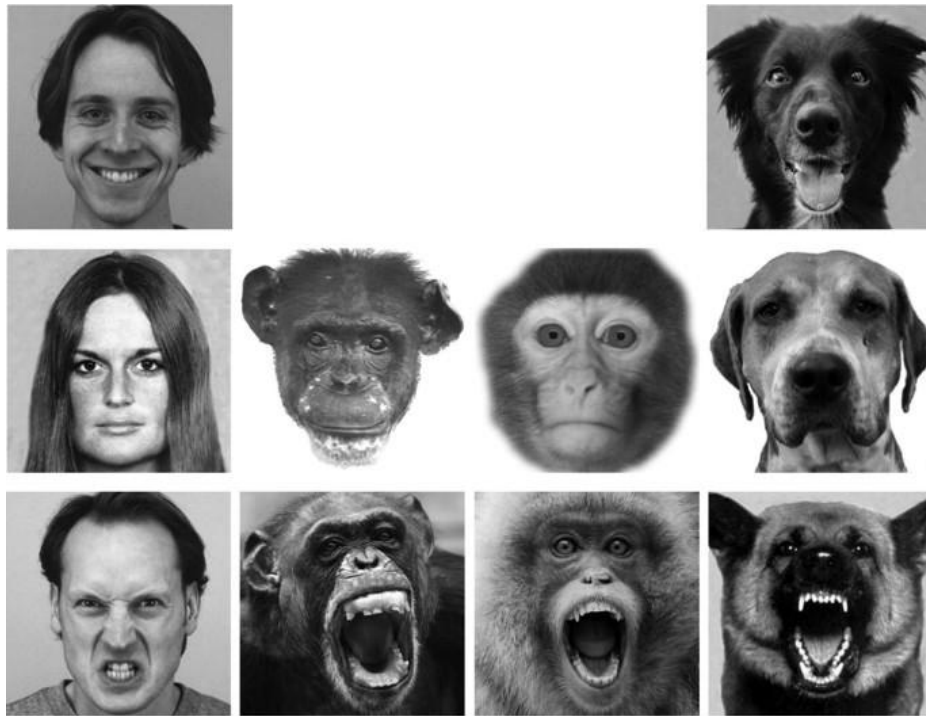
Learn From ^{Data} Experience



Follow Instructions

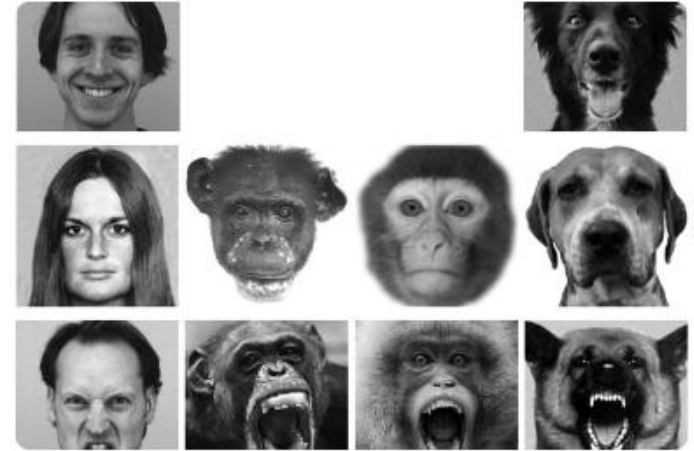


Manusia belajar dari pengalaman, kita dapat memberikan kemampuan (*ability*) ke *computer* agar dapat belajar dari data.



Menurut anda, AI vision mana yang lebih bagus dalam menganalisis gambar di atas, OpenAI GPT-4o atau Google Gemini 1.5 Flash?

OpenAI GPT-4o (Vision Capability)



Apakah semua ada gambar manusia? hewan apa saja yang ditampilkan pada gambar?



Pada gambar ini, terdapat campuran gambar manusia dan hewan.

- **Gambar manusia:** Terdapat tiga gambar manusia, yaitu gambar di kolom pertama (baris pertama dan kedua) serta kolom ketiga (baris ketiga).
- **Gambar hewan:** Gambar hewan yang ditampilkan adalah:
 - **Simpanse:** Terdapat pada kolom kedua di baris kedua dan ketiga.
 - **Monyet:** Terdapat pada kolom ketiga di baris kedua dan ketiga.
 - **Anjing:** Terdapat pada kolom keempat di baris pertama, kedua, dan ketiga.

Jadi, dalam gambar ini ada campuran antara manusia, simpanse, monyet, dan anjing.

Google Gemini 1.5 Flash Vision

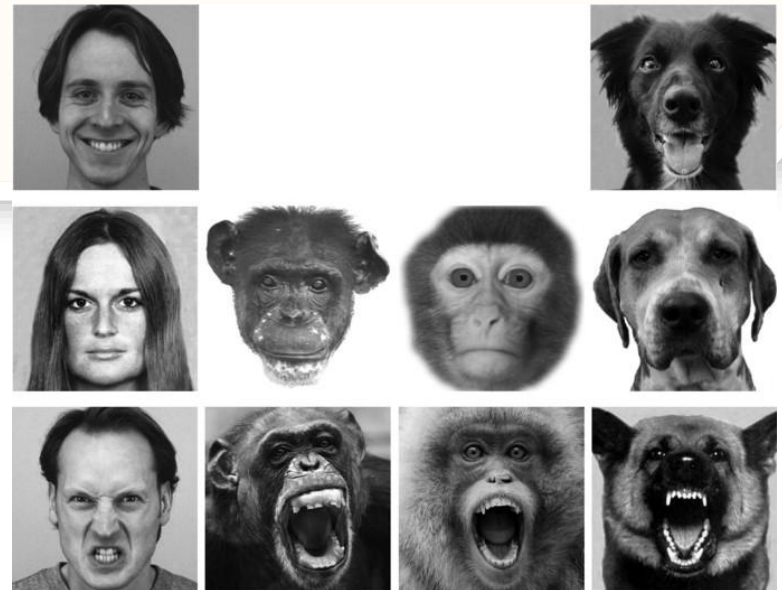


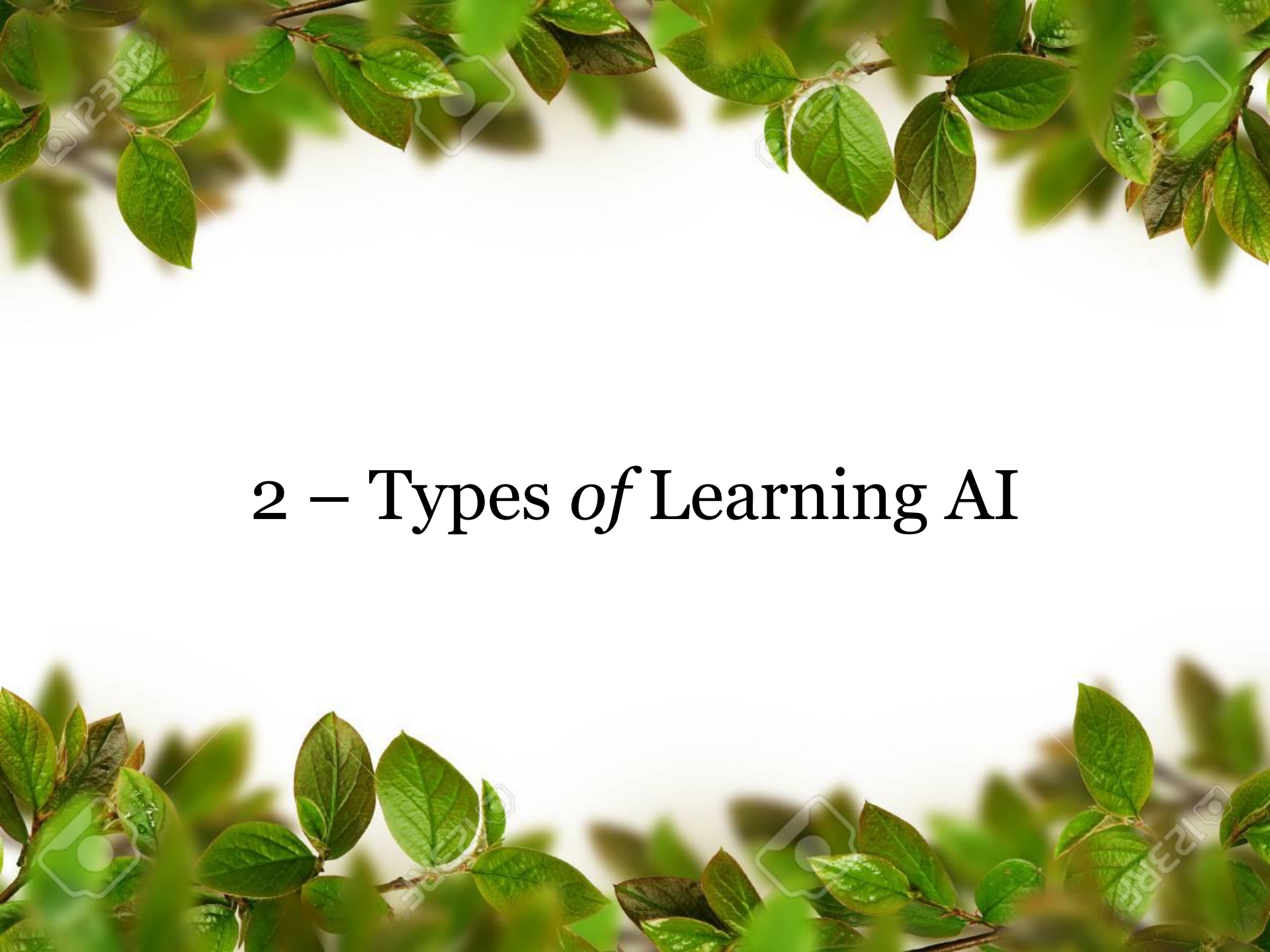
```
5 $ImagePath = "C:/Users/USER/Downloads/monkey.jpg";  
6 // Encode gambar ke base64  
7 $ImageData = base64_encode(file_get_contents($ImagePath));  
8  
9 // URL endpoint API  
10 $url = "https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?key={$apiKey}";  
11  
12 // Data JSON yang akan dikirim  
13 $data = [  
14     "contents" => [  
15         [  
16             "parts" => [  
17                 ["text" => "Apakah semua ada gambar manusia? hewan apa saja yang ditampilkan pada gambar?"] ,  
18                 ["inline_data" => [  
19                     "mime_type" => "image/png",  
20                     "data" => $ImageData  
21                 ] ]  
22             ]  
23         ]  
24     ]  
25 ];
```

"Output":

Gambar tersebut menunjukkan **tiga wajah manusia** dan **enam hewan**. Hewan-hewan tersebut adalah:

- dua simpanse
- dua monyet
- satu anjing berbulu hitam
- satu anjing berbulu coklat.



The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some brownish edges, suggesting a natural, organic theme.

2 – Types of Learning AI

Penemuan Listrik

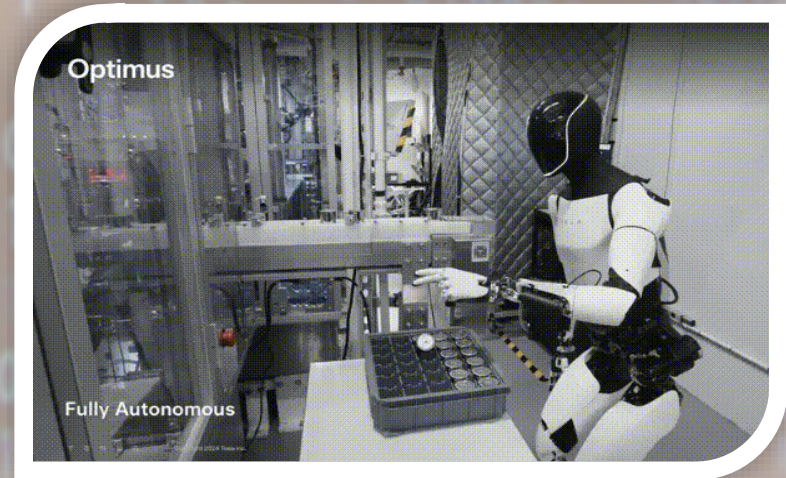


"Artificial intelligence is the new electricity."

— Andrew Ng (Pakar AI & Deep Learning)

Ketika listrik pertama kali diperkenalkan pada abad ke-18 (1701-1800) dan abad ke-19 (tahun 1801-1900), listrik dianggap misterius dan berbahaya. Saat Thomas Edison memperkenalkan lampu pijar, masyarakat lebih percaya pada lilin dan lampu minyak. Namun, listrik terbukti lebih aman dan meningkatkan produktivitas, mendorong Revolusi Industri. Seiring waktu, masyarakat mulai menerima teknologi ini setelah merasakan manfaatnya.

Teknologi AI



Teknologi AI juga dianggap berisiko dan diragukan. Pada **1997**, Deep Blue mengejutkan dunia dengan mengalahkan juara catur Garry Kasparov, membuktikan bahwa mesin bisa melampaui manusia dalam tugas tertentu. Pada **2016**, AlphaGo dari DeepMind kembali mencetak sejarah dengan menaklukkan juara dunia Go game, Lee Sedol, menggunakan strategi yang bahkan belum pernah dipikirkan manusia. Meskipun awalnya banyak yang skeptis/ragu, teknologi AI mulai membuktikan manfaatnya di berbagai **bidang (kesehatan & industri)**. Seiring waktu, teknologi ini bisa menjadi pilar inovasi di era modern.

SKILL AI
bukan
NILAI TAMBAH



DeepSeek
(AI Industry R&D)



5G Networks
(Huawei 5G Technology)



Robot Installations
(Industrial Robotics Market)

Skill AI bukan lagi sekadar nilai tambah, tetapi sudah menjadi kebutuhan di era digital saat ini.

SKILL AI
jadi
KEBUTUHAN



Commercial Drones
(Drones Industry)

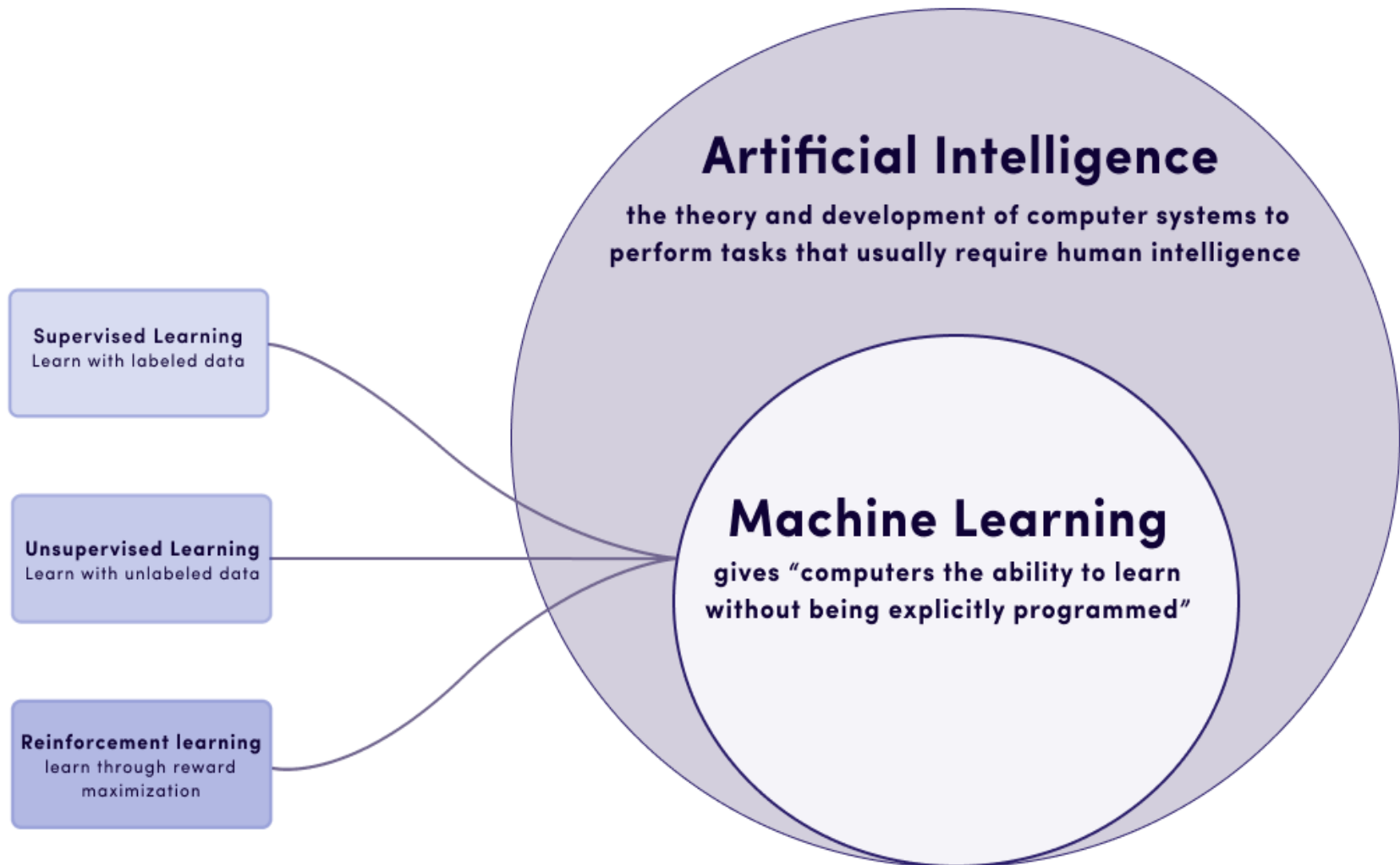


TikTok
(Largest Social Media Market)

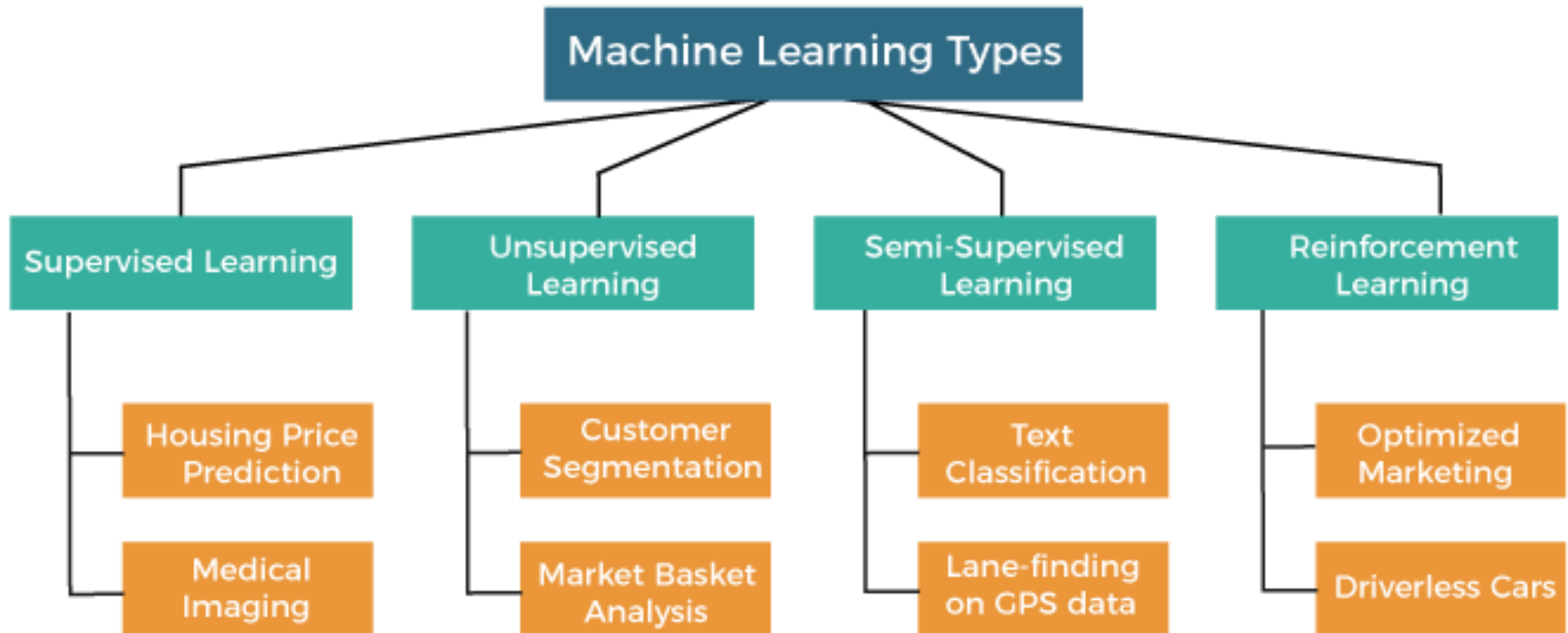


Autonomous Cars
(Self-driving Car Industry)

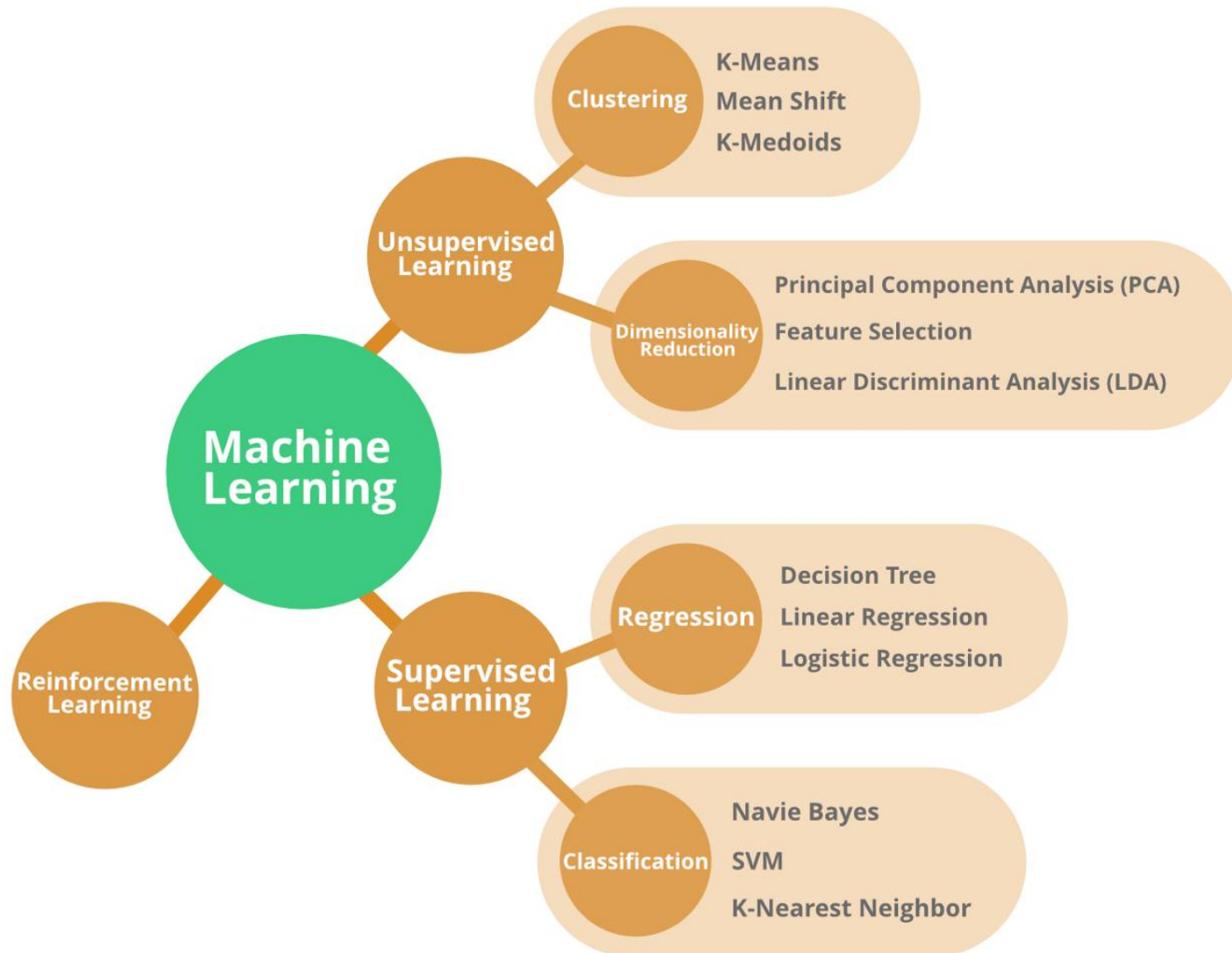
Different Types of Machine Learning



Types of Learning AI



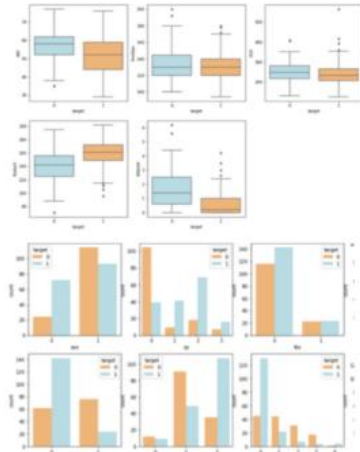
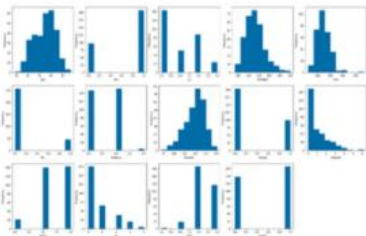
Types of Learning AI



Machine Learning Algorithms - Classification

Exploratory Data Analysis (EDA)

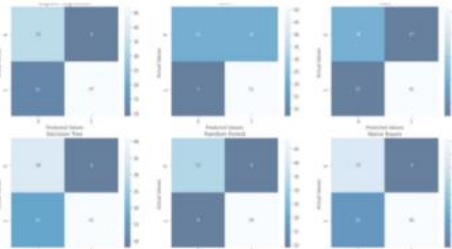
- 1) **Histogram:** `df.plot(kind = 'hist')`
- 2) **Box Plot:** `sns.boxplot()`
- 3) **Grouped Bar Chart:** `sns.countplot()`



Model Evaluation

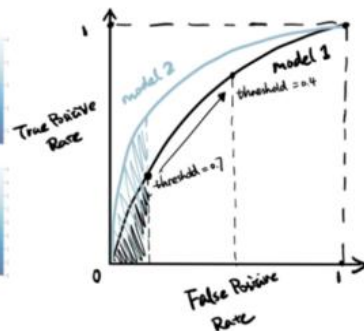
Confusion Matrix

`confusion_matrix(y_test, y_pred)`

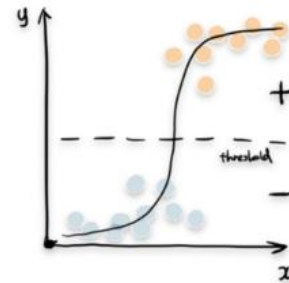


ROC & AUC

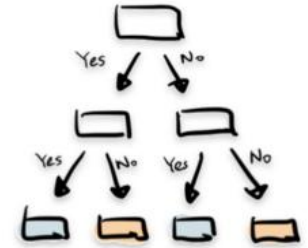
`metrics.auc(fpr, tpr)`



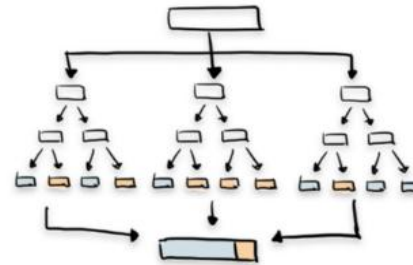
Logistic Regression



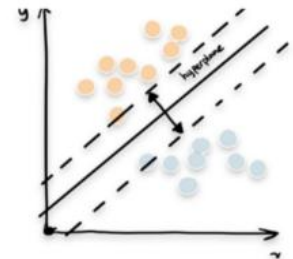
Decision Tree



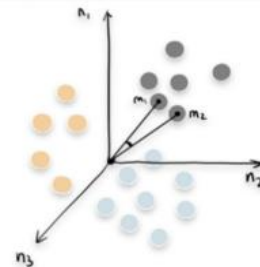
Random Forest



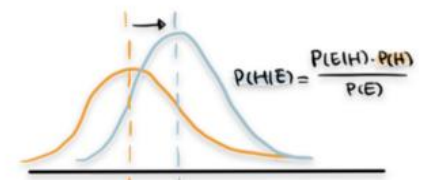
Support Vector Machine



K Nearest Neighbour

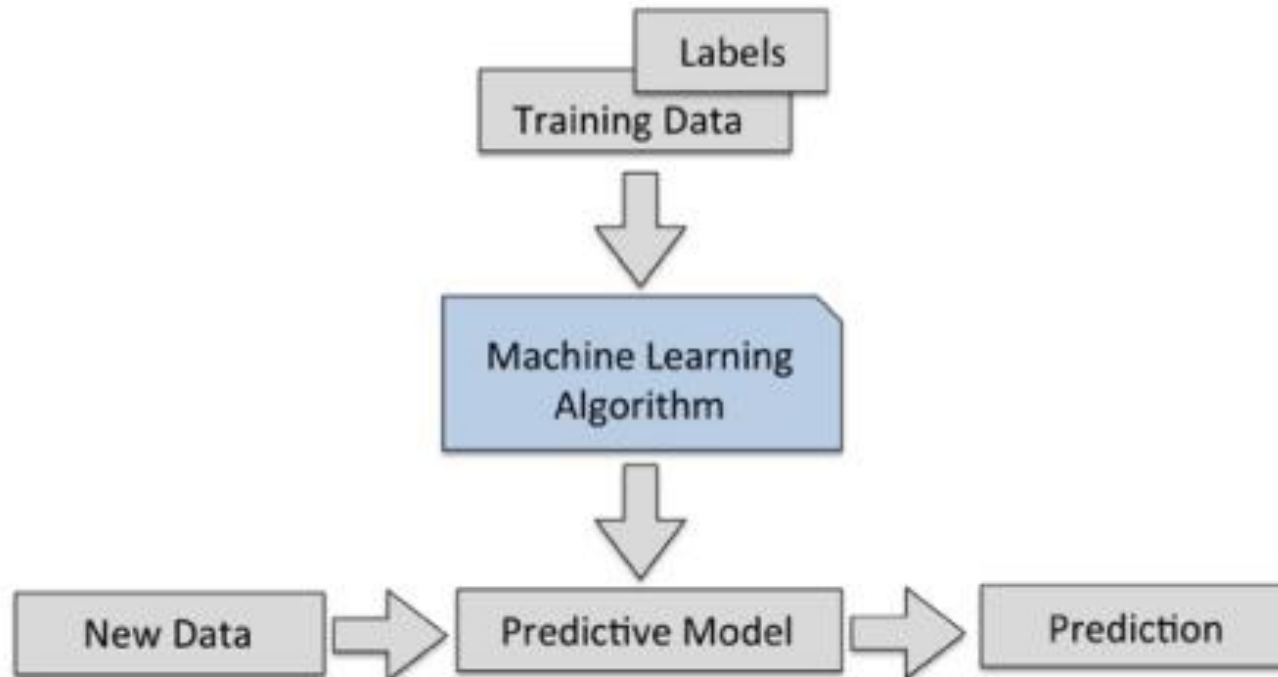


Naive Bayes



Making Predictions

Making predictions about the future with supervised learning.

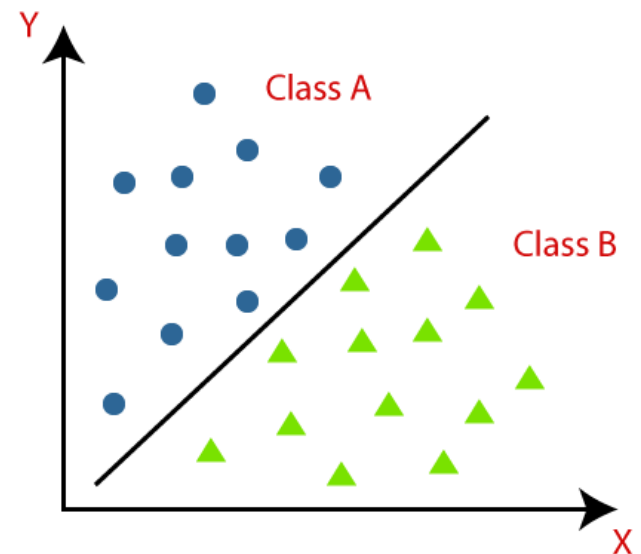
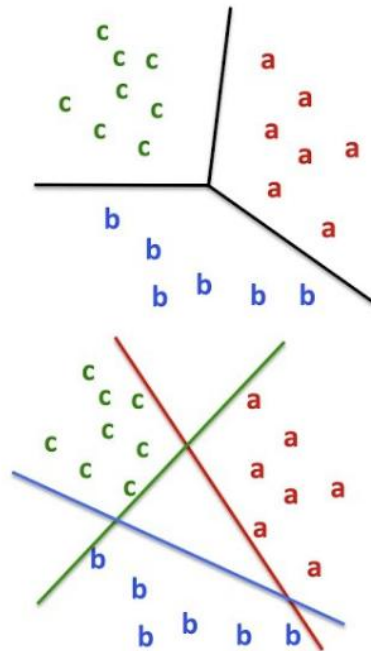


The main goal in supervised learning is to learn a model from **labeled training data** that allows us to make predictions about unseen or future data. Here, the term supervised refers to a set of samples where the desired output signals (*labels*) are already known.

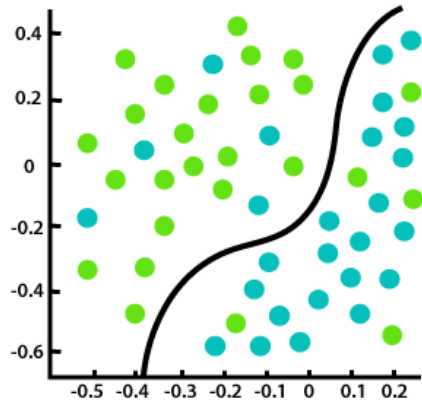
Classification *for* predicting class labels

- ❖ Classification is a subcategory of supervised learning where the goal is to **predict the categorical class labels** of new instances based on past observations.
- ❖ Those class labels are discrete, unordered values that can be understood as the group memberships of the instances.
- ❖ The previously mentioned example of e-mail-spam detection represents a typical example of a **binary classification task**, where the machine learning algorithm learns a set of rules in order to distinguish between two possible classes: spam and non-spam e-mail.

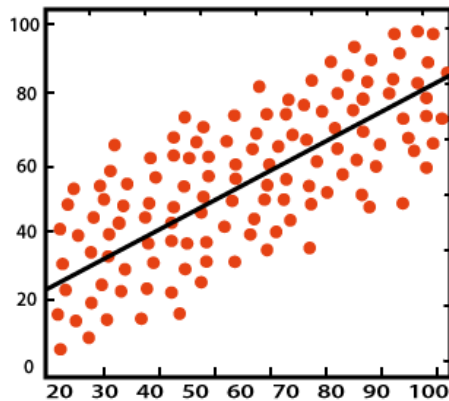
- Multi-class:
 - classes mutually exclusive:
 - instance is either a or b or c
 - even if it's an outlier
 - NB, kNN, DT, logistic
- Binary:
 - one-vs-rest:
 - {a} vs {not a}, {b} vs {not b}
 - classes may overlap
 - instance can be both a and b
 - can be in none of the classes
 - SVM, logistic, perceptron



Classification *vs* Regression *vs* Clustering



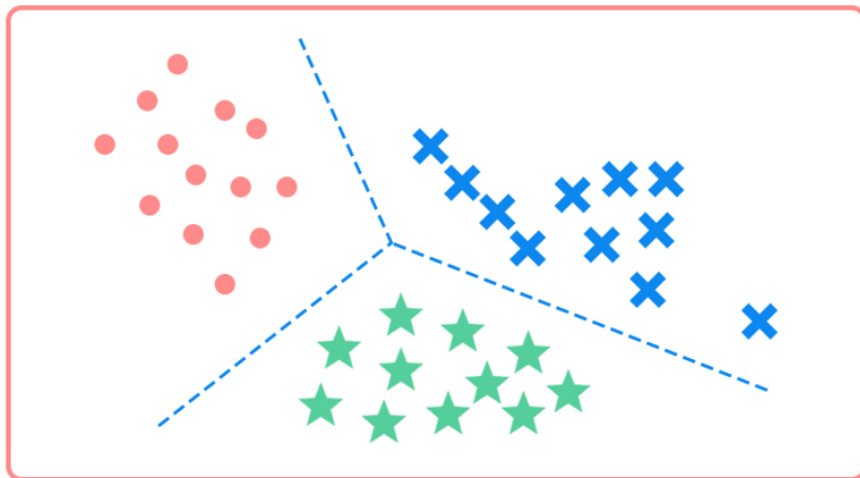
Classification



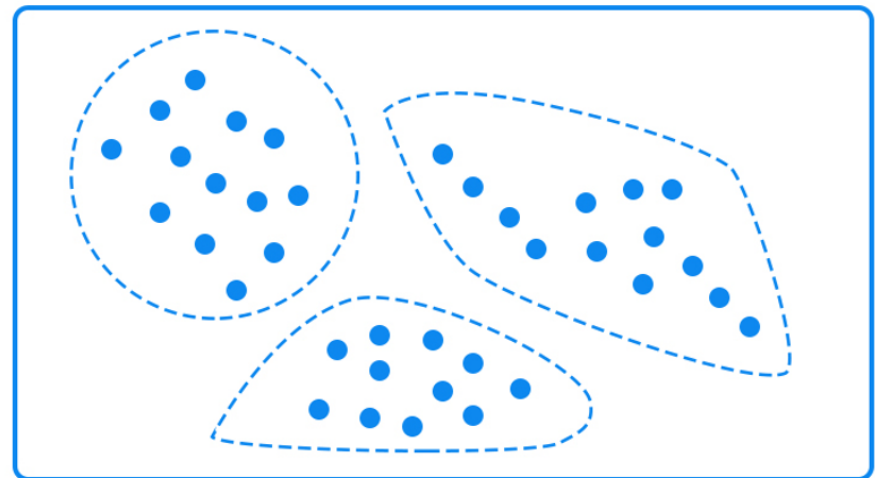
Regression

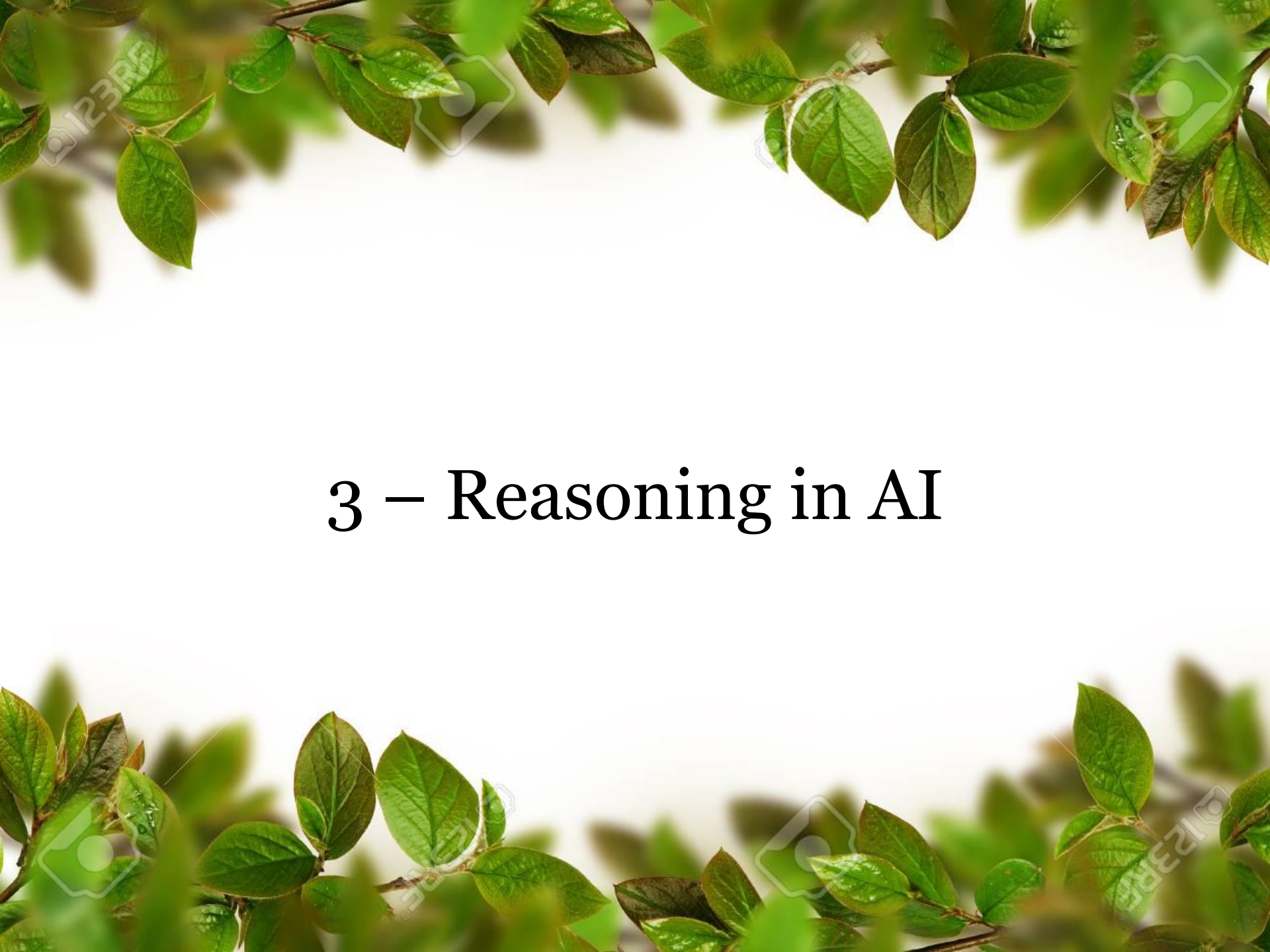
The most significant difference between regression and classification is that while regression helps predict a continuous quantity, classification predicts discrete class labels. There are also some overlaps between the two types of machine learning algorithms.

Classification



Clustering



The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some brownish edges, suggesting a natural, organic theme.

3 – Reasoning in AI

Dapatkah AI
sekarang ini
berpikir sebelum
memberikan
respons?

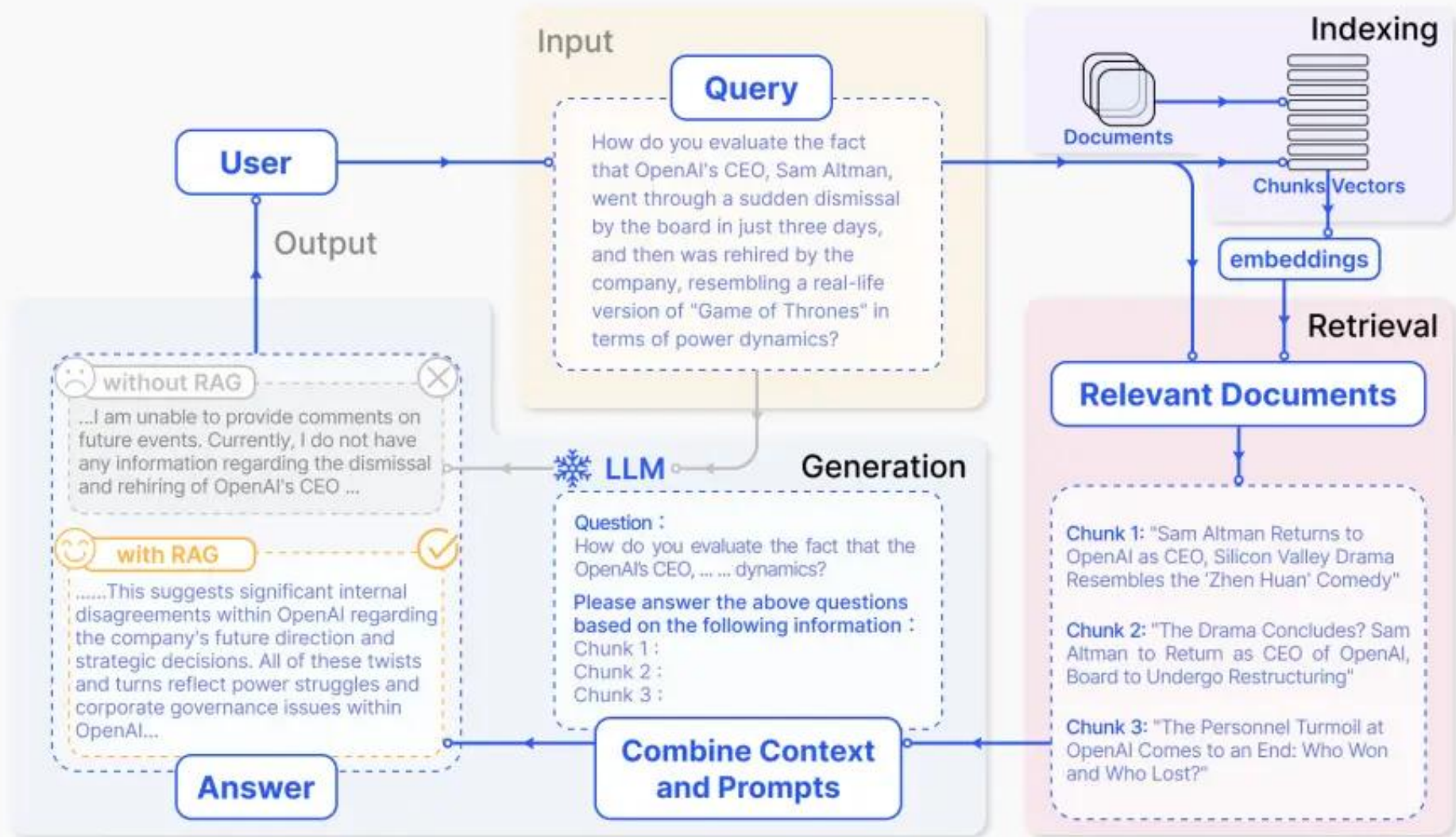
What is Reasoning?

III Knowledge, reasoning, and planning

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<https://www.moveworks.com/us/en/resources/ai-terms-glossary/reasoning>

- ❖ **AI reasoning** is how **artificial intelligence systems solve problems, think critically, and create new knowledge** by **analyzing and processing available information**.
- ❖ **Reasoning** allows AI systems to make logical inferences and **draw conclusions from available data**, moving **beyond basic understanding to more advanced decision-making**.
- ❖ Foundation models like **LARGE LANGUAGE MODELS** provide a strong base for **reasoning**. These models are trained on vast/large amounts of text and can interpret and generate human-like language.
- ❖ However, the **foundation model alone is not enough for complex reasoning across different tasks**. The model must be fine-tuned and extended with additional capabilities.
- ❖ This involves techniques like adjusting the model's parameters, adding specialized model layers, incorporating knowledge bases, and more. These extensions build upon the foundation to impart **domain-specific reasoning abilities**.



Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., ... & Wang, H. (2023). **Retrieval-augmented generation for large language models: A survey**. arXiv preprint arXiv:2312.10997.

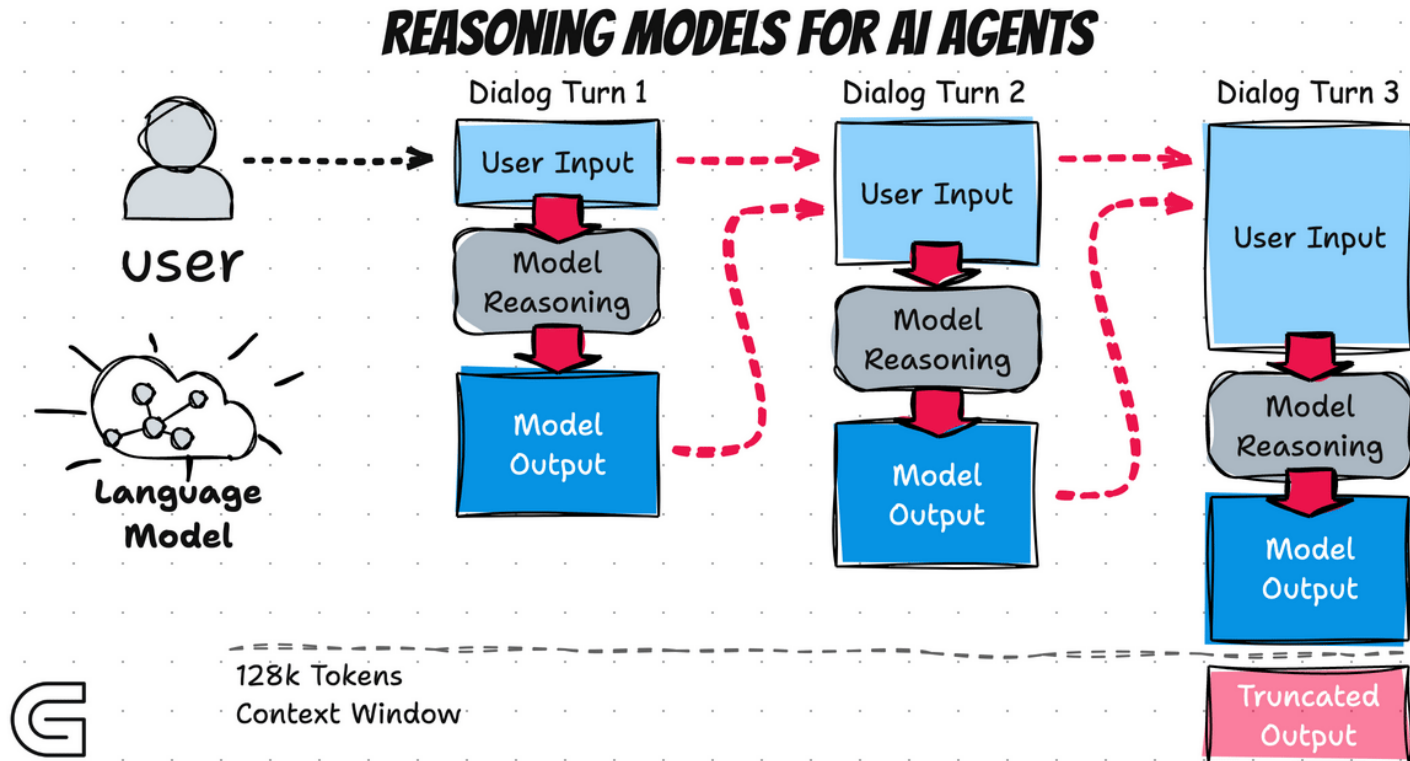
AI Reasoning & LLMs

AI reasoning dan **Large Language Models (LLMs)** memiliki hubungan yang erat karena LLMs merupakan salah satu bentuk AI yang memiliki kemampuan reasoning (*penalaran*) dalam berbagai tingkat.

AI reasoning is an AI system's ability to analyze, understand and make decisions based on available information. In LLMs, reasoning plays a role in:

Logical reasoning	Understanding cause-and-effect relationships, making logical inferences.
Commonsense reasoning	Using real-world knowledge to infer things.
Mathematical reasoning	Solving math problems or thinking quantitatively.
Analogical reasoning	Drawing conclusions based on similar patterns.

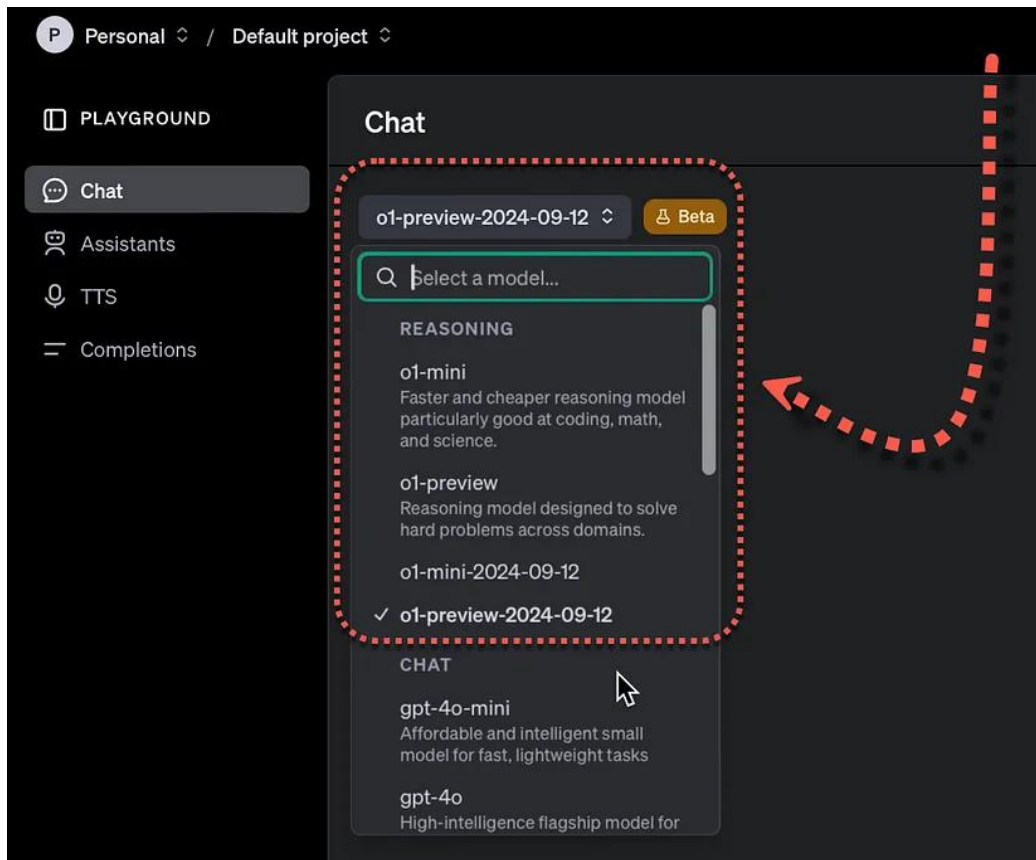
Deep Learning Architectures



Although, LLMs are mainly statistical and probabilistic based (using deep learning to predict the next word), they have developed some form of reasoning through large-scale training.

OpenAI o1 Reasoning Models

Reasoning models can “**think**” before responding, generating a detailed internal chain of thought that simulates human-like deliberation. This process allows them to break down complex problems step-by-step, improving both accuracy and depth in their responses.



<https://cobusgreyling.medium.com/openai-o1-reasoning-models-49ccff5b3a20>

OpenAI Imagines Our AI Future

Stages of Artificial Intelligence

Level 1	Chatbots, AI with conversational language
Level 2	Reasoners, human-level problem solving
Level 3	Agents, systems that can take actions
Level 4	Innovators, AI that can aid in invention
Level 5	Organizations, AI that can do the work of an organization

OpenAI has unveiled its plan for generalizing its technology, which can be interpreted as a road map, a statement of intent, or a way to maintain momentum. There are no big surprises, as it aligns with what was expected to happen; we understand it as a confirmed standardization.

OpenAI Reasoning Models

Reasoning models for complex, multi-step problems:

<https://openai.com/api/pricing/>

Reasoning models for complex, multi-step problems

OpenAI o1

Frontier reasoning model that supports tools, Structured Outputs, and vision | 200k context length

Price

Input:

\$15.00 / 1M tokens

Cached input:

\$7.50 / 1M tokens

Output:

\$60.00 / 1M tokens

OpenAI o3-mini

Small cost-efficient reasoning model that's optimized for coding, math, and science, and supports tools and Structured Outputs | 200k context length

Price

Input:

\$1.10 / 1M tokens

Cached input:

\$0.55 / 1M tokens

Output:

\$4.40 / 1M tokens

OpenAI Reasoning Models

The screenshot shows the ChatGPT web interface. The browser tab is labeled 'ChatGPT' and the address bar shows the URL 'chatgpt.com/c/67b53686-942c-8006-9c84-b906fc94fea9'. The left sidebar contains a search icon, a 'ChatGPT' dropdown, and a 'Today' section with a 'New chat' button. The main chat area displays a user prompt: 'Siapa presiden Indonesia di tahun 2025?'. Below the prompt, the model's response is shown, starting with 'Considering Indonesian President in 2025 >'. The response text is: 'The Indonesian election happened in 2024, so Jokowi (Joko Widodo) was still the president until then, as his second term ends in 2024. Based on the constitution, he couldn't run again. By 2025, a new president might have been sworn in, possibly someone like Ganjar Pranowo if he won the election. But given my data cutoff, I can't confirm who the new president is for sure. I can only say Jokowi was president until 2024.' At the bottom, there is a 'Message ChatGPT' input field with a 'Think before responding' tooltip, a 'Reason' button, and a 'Search' button. A footer at the bottom right states 'ChatGPT can make mistakes. Check important info.'

ChatGPT

chatgpt.com/c/67b53686-942c-8006-9c84-b906fc94fea9

ChatGPT

Explore GPTs

Today

New chat

Siapa presiden Indonesia di tahun 2025?

Considering Indonesian President in 2025 >

The Indonesian election happened in 2024, so Jokowi (Joko Widodo) was still the president until then, as his second term ends in 2024. Based on the constitution, he couldn't run again. By 2025, a new president might have been sworn in, possibly someone like Ganjar Pranowo if he won the election. But given my data cutoff, I can't confirm who the new president is for sure. I can only say Jokowi was president until 2024.

Message ChatGPT

Think before responding

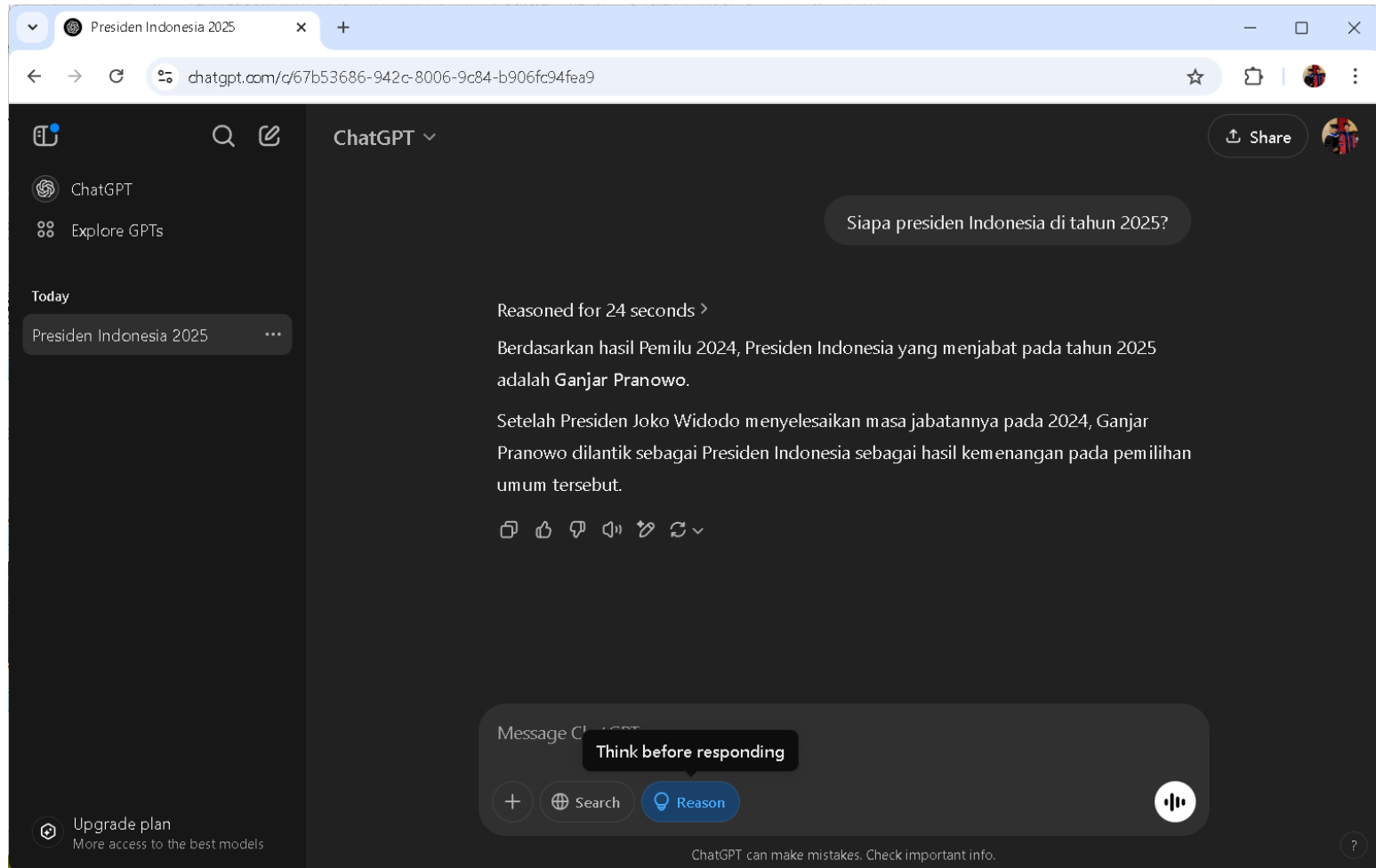
Reason

Search

Upgrade plan
More access to the best models

ChatGPT can make mistakes. Check important info.

OpenAI Reasoning Models



Mengapa nama
presiden Indonesia
tahun 2025 tidak
diketahui?

LLM Model Overview Input/Output Tokens (docsbot.ai)

Feature	 GPT-4o	 Gemini 2.0 Flash
Input Context Window The number of tokens supported by the input context window.	128K tokens	1M tokens
Maximum Output Tokens The number of tokens that can be generated by the model in a single request.	16.4K tokens	8,192 tokens
Open Source Whether the model's code is available for public use.	No	No
Release Date When the model was first released.	August 6, 2024 6 months ago	December 11, 2024 2 months ago
Knowledge Cut-off Date When the model's knowledge was last updated.	October 2023	August 2024
API Providers The providers that offer this model. (This is not an exhaustive list.)	OpenAI, Azure OpenAI Service	Google AI Studio, Vertex AI

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some brownish edges, suggesting a natural, organic theme.

3 – Logical Agents

Why is reasoning important?

III Knowledge, reasoning, and planning

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- ❖ Reasoning is a crucial capability that allows AI systems to make logical inferences and come to conclusions based on available information. It builds upon foundation models to move beyond basic understanding toward advanced decision-making.
- ❖ Reasoning empowers AI to draw connections between disparate data, make nuanced judgments in complex domains, predict hypothetical scenarios, weigh alternatives, and determine actions. This ability to derive insights is key to developing AI that can assist human experts and enhance processes spanning many industries.

Reasoning Dalam Artificial Intelligence

Reasoning (penalaran) adalah proses mental untuk memperoleh kesimpulan logis dan membuat prediksi dari pengetahuan, fakta, dan kepercayaan yang tersedia. Atau kita bisa mengatakan, "Penalaran adalah cara menyimpulkan fakta dari data yang ada." Ini adalah proses umum berpikir secara rasional, untuk menemukan kesimpulan yang valid.

Dalam kecerdasan buatan, penalaran sangat penting agar mesin juga dapat berpikir secara rasional seperti otak manusia, dan dapat bekerja seperti manusia. Penalaran pun dalam artificial intelligence dibagi ke dalam beberapa jenis, diantaranya:

- Deductive reasoning
- Inductive reasoning
- Abductive reasoning
- Common Sense Reasoning
- Monotonic Reasoning
- Non-monotonic Reasoning

Tipe-Tipe Reasoning Artificial Intelligence

Deductive Reasoning

Penalaran deduktif menyimpulkan informasi baru dari informasi yang diketahui terkait secara logis. Ini adalah bentuk penalaran yang valid, yang berarti kesimpulan argumen harus benar jika premisnya benar.

Penalaran deduktif adalah jenis logika proposisional dalam AI, dan membutuhkan berbagai aturan dan fakta. Kadang-kadang disebut sebagai penalaran top-down, dan bertentangan dengan penalaran induktif.

7

LOGICAL AGENTS

In which we design agents that can form representations of a complex world, use a process of inference to derive new representations about the world, and use these new representations to deduce what to do.

REASONING
REPRESENTATION
KNOWLEDGE-BASED
AGENTS

Humans, it seems, know things; and what they know helps them do things. These are not empty statements. They make strong claims about how the intelligence of humans is achieved—not by purely reflex mechanisms but by processes of **reasoning** that operate on internal **representations** of knowledge. In AI, this approach to intelligence is embodied in **knowledge-based agents**.

The problem-solving agents of Chapters 3 and 4 know things, but only in a very limited, inflexible sense. For example, the transition model for the 8-puzzle—knowledge of what the actions do—is hidden inside the domain-specific code of the `RESULT` function. It can be used to predict the outcome of actions but not to deduce that two tiles cannot occupy the same space or that states with odd parity cannot be reached from states with even parity. The atomic representations used by problem-solving agents are also very limiting. In a partially observable environment, an agent's only choice for representing what it knows about the

7.1 KNOWLEDGE-BASED AGENTS

KNOWLEDGE BASE

SENTENCE

KNOWLEDGE
REPRESENTATION
LANGUAGE
AXIOM

The central component of a knowledge-based agent is its **knowledge base**, or KB. A knowledge base is a set of **sentences**. (Here “sentence” is used as a technical term. It is related but not identical to the sentences of English and other natural languages.) Each sentence is expressed in a language called a **knowledge representation language** and represents some assertion about the world. Sometimes we dignify a sentence with the name **axiom**, when the sentence is taken as given without being derived from other sentences.

INFERENCE

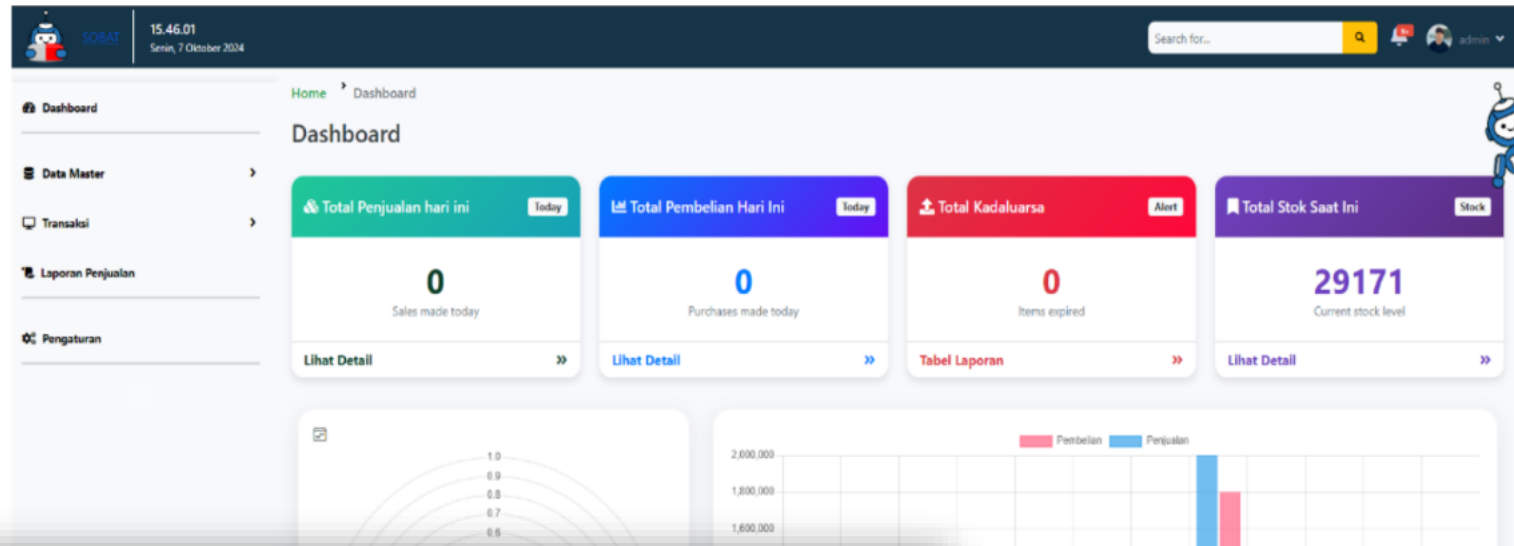
There must be a way to add new sentences to the knowledge base and a way to query what is known. The standard names for these operations are TELL and ASK, respectively. Both operations may involve **inference**—that is, deriving new sentences from old. Inference must obey the requirement that when one ASKs a question of the knowledge base, the answer should follow from what has been told (or TELLED) to the knowledge base previously. Later in this chapter, we will be more precise about the crucial word “follow.” For now, take it to mean that the inference process should not make things up as it goes along.

BACKGROUND
KNOWLEDGE

Figure 7.1 shows the outline of a knowledge-based agent program. Like all our agents, it takes a percept as input and returns an action. The agent maintains a knowledge base, *KB*, which may initially contain some **background knowledge**.

Role-Based Agents act based on predefined roles with fixed rules or scenarios. Knowledge-Based Agents use knowledge and logic to understand the environment and make more flexible decisions. The main difference is that Role-Based Agents are more static, while Knowledge-Based Agents are more adaptive and can learn from new information.

Skripsi Fakultas Ilmu Komputer



The Med-Bot chat interface shows a user asking for a list of cough medicines. The bot responds with a list of available medicines, including their codes, units, prices, and stock levels.

Search Drug
Search drug...

Nama Obat

Harga
Rp

Jumlah

Total Harga

Med-Bot

Berikut adalah daftar obat batuk yang tersedia

No	Nama Obat	Kode Obat	Satuan	Harga	Stok	Indikasi
1.	ACTIFED PLUS EXPECTORANT- HIJAU	ACTIFEDEXPECT	Sirup (syr)	Rp 25.000,00	5	Batuk berdahak
2.	ACTIFED PLUS COUGH SYRUP 60 ML - MERAH					

Ask your question here...

Send

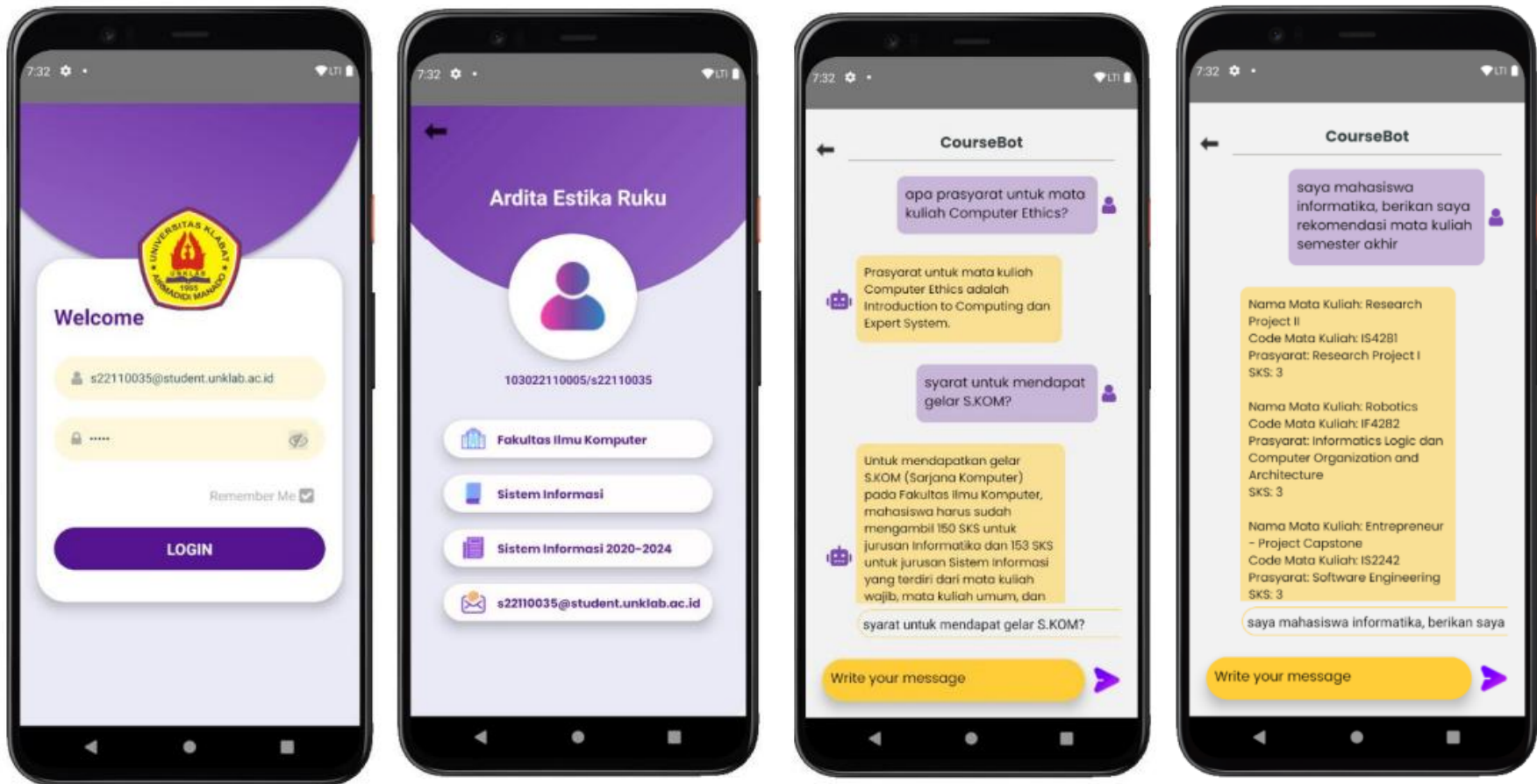
Medicine Inventory
Management
(LLM+MySQL+WhatsApp Gateway)

Skripsi Fakultas Ilmu Komputer



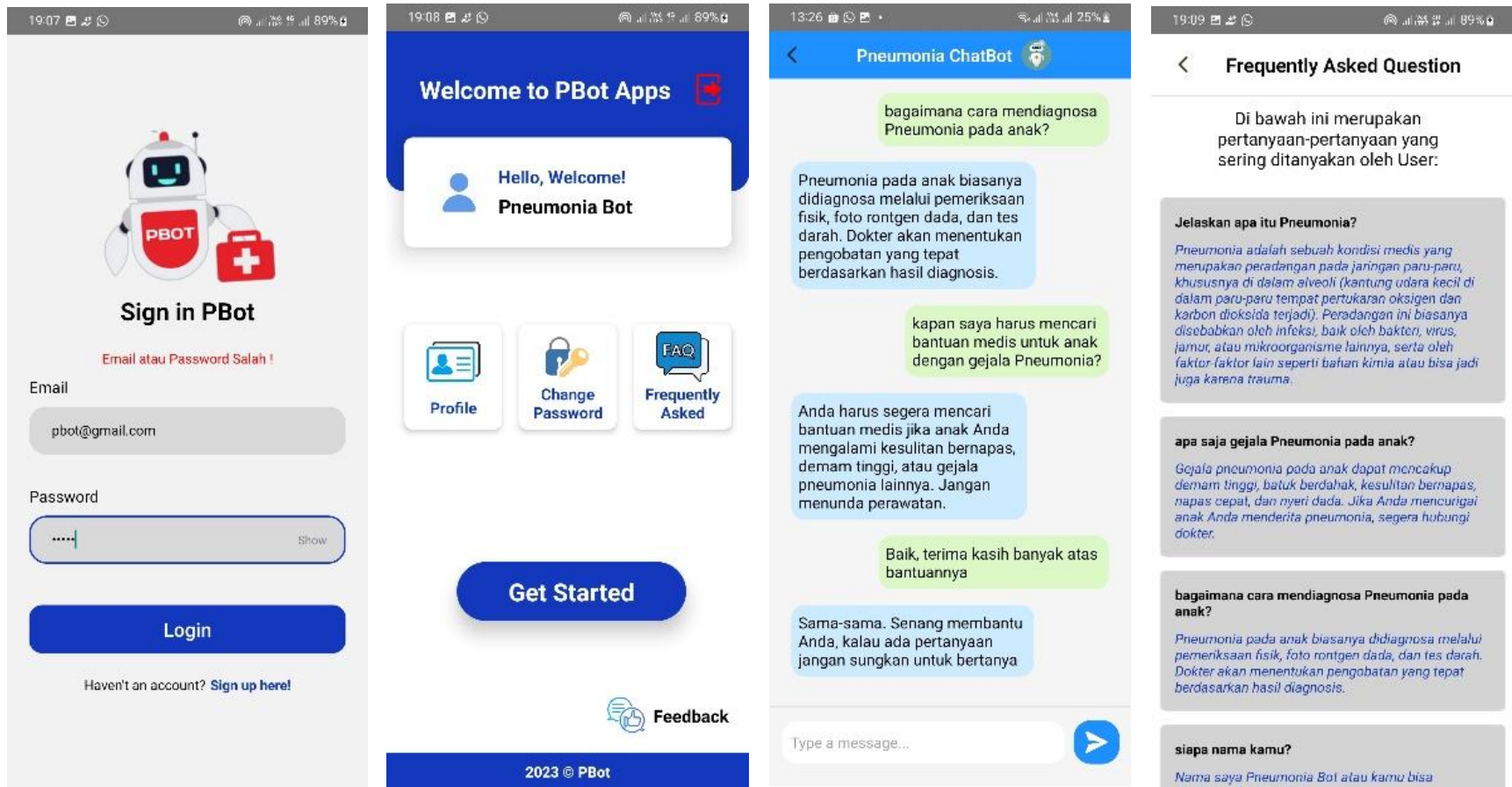
Health & Nutrition Information Provider
Chatbot (LLM+RAG)

Skripsi Fakultas Ilmu Komputer



Course Recommendation Chatbot
(React+LLM+RAG)

Skripsi Fakultas Ilmu Komputer



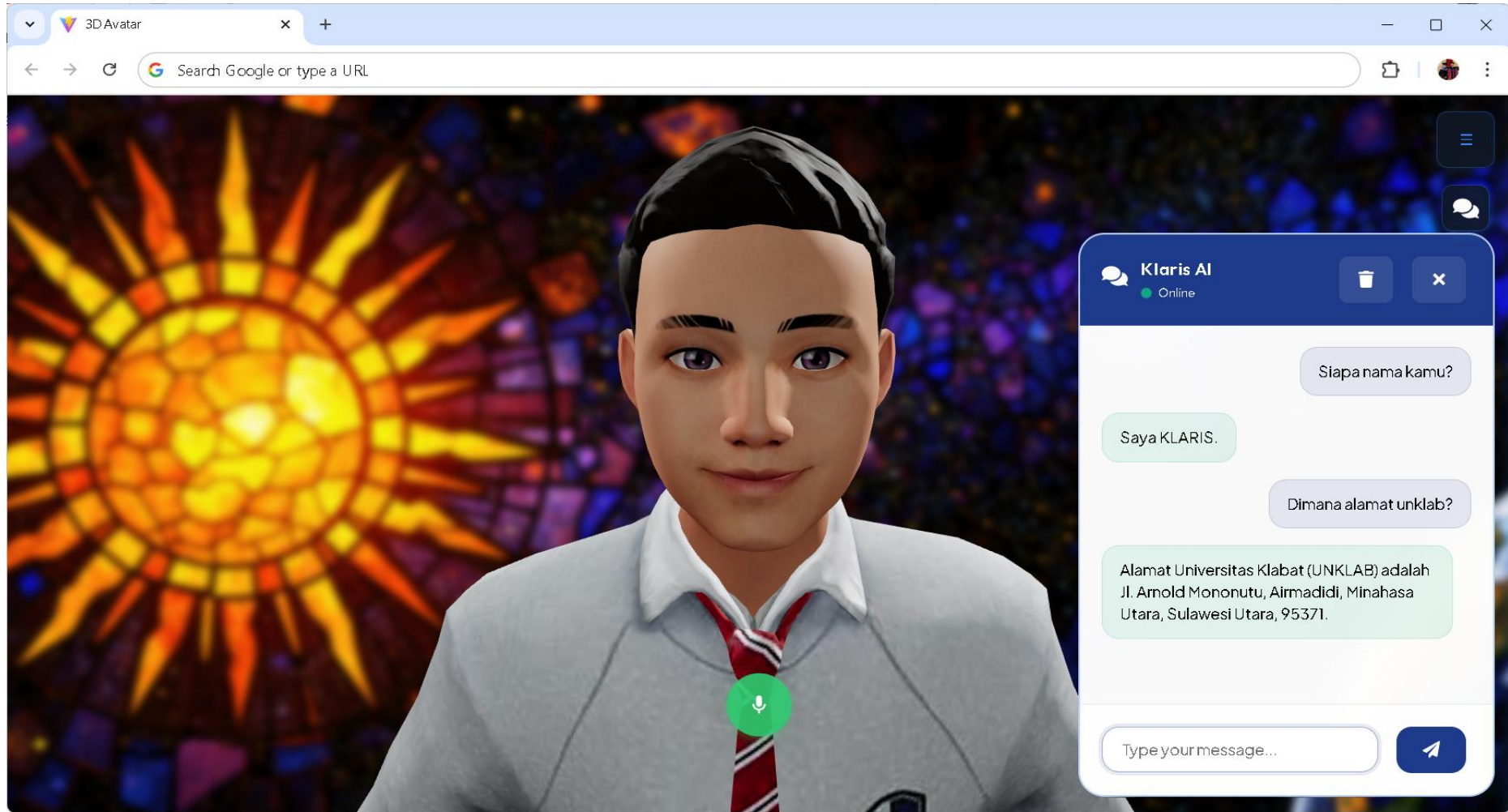
Child Pneumonia Diagnosis Chatbot (AIML+LLM)

Skripsi Fakultas Ilmu Komputer



Be Alert to Report Disasters –
(LLM+VLM+WhatsApp Gateway)

Skripsi Fakultas Ilmu Komputer



Klaris (AI Unklab) – 3D+TTS+TTS+LLM+RAG



Exercise *for* Students

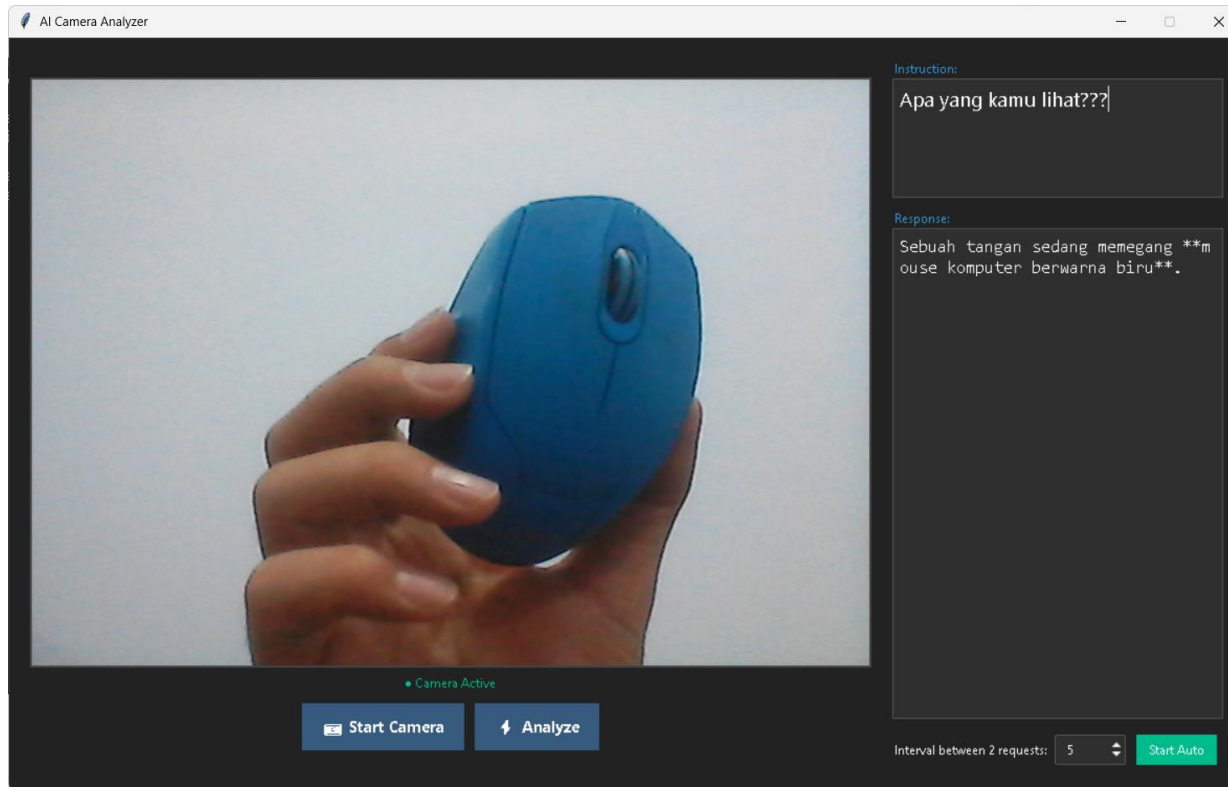
CLASS ACTIVITY #1

(Vision Capability in Large Language Model (LLM))

Mendemokan vision capability dari Large Language Model (LLM) untuk mendeteksi Object Detection pada gambar

2026

Activity Overview



- 1) Pada activity ini, students akan melakukan demo kemampuan vision (visual understanding) dari sebuah Large Language Model (LLM).
- 2) The main goal adalah menguji apakah LLM dapat mengenali dan mendeskripsikan objects yang terdapat di dalam sebuah image.
- 3) Students akan memberikan sebuah gambar kepada LLM yang memiliki kemampuan vision, kemudian meminta model untuk melakukan Object Detection / Object Identification terhadap gambar tersebut.

Learning Objectives

Setelah menyelesaikan activity ini, students diharapkan mampu:

1. Memahami konsep dasar **Computer Vision** dan **Vision-Language Model (VLM)**.
2. Memahami bagaimana **Large Language Model with Vision Capability** dapat menganalisis sebuah image.
3. Melakukan **Object Detection** menggunakan image sebagai input.
4. Membuat prompt yang efektif untuk mendapatkan informasi object dari sebuah image.
5. Menganalisis dan mengevaluasi hasil detection dari AI.
6. Memahami **limitations** (kekurangan dan batasan) dan kemungkinan error pada AI vision system.

Verification - Human vs AI

Setelah mendapatkan hasil dari AI, students harus melakukan manual verification. Bandingkan antara “**Actual Objects in the Image**” dan “**Objects Detected by the LLM**”.

Students harus mengidentifikasi:

Correct Detection

Object benar-benar ada dan berhasil dikenali AI.

Missed Detection

Object ada tetapi tidak dikenali AI.

False Detection

AI mendeteksi object yang sebenarnya tidak ada.

Misclassification

Object ada tetapi AI memberikan nama/class yang salah.

Analysis Questions

Jawab pertanyaan berikut berdasarkan hasil experiment:

1. How many **objects were detected by the LLM**?
2. Apakah semua objects dalam image berhasil dideteksi?
3. Apakah ada **Missed Detection**?
4. Apakah terdapat **False Detection**?
5. Apakah ada object yang mengalami **Misclassification**?
6. Object apa yang paling mudah dikenali oleh AI? Why?
7. Object apa yang paling sulit dikenali oleh AI? Why?
8. Bagaimana kualitas image memengaruhi hasil detection?
9. Bagaimana perubahan prompt memengaruhi hasil dari LLM?
10. Menurut hasil experiment, apa saja **limitations of LLM Vision Capability**?

Discussion

Students harus membuat kesimpulan singkat mengenai:

"How capable is an LLM in understanding and detecting objects in an image?"

Dalam conclusion, students harus menjelaskan berdasarkan hasil experiment, bukan hanya berdasarkan asumsi.

Gemini Developer API Pricing 2026

Link Access: <https://aistudio.google.com/docs/pricing>

Google AI Studio

< Documentation

Filter docs...

API reference ↗

Community ↗

GET STARTED

Overview

Get started

API keys

Pricing

Coding agent setup

MODELS

All models

Latest Gemini models

staju1989@gmail.com

Gemini 3.8 Flash

`gemini-3.8-flash` Try it in Google AI Studio

Our most intelligent Flash model, engineered for long-horizon software engineering, autonomous agents, and complex enterprise workflows.

	Standard	Batch	Flex	Priority
Feature	Free Tier			Paid Tier, per 1M tokens in USD
Input price	Free of charge			\$0.75 through December 31, 2026. \$1.50 starting January 1, 2027.
Output price (including thinking tokens)	Free of charge			\$3.75 through December 31, 2026. \$7.50 starting January 1, 2027.
Context caching price	Free of charge			\$0.075 through December 31, 2026. \$0.15 starting January 1, 2027. \$0.50 / 1,000,000 tokens per hour (storage price) through December 31, 2026. \$1.00 / 1,000,000 tokens per hour (storage price) starting

ON THIS PAGE

- Gemini 3.8 Flash
- Gemini 3.7 Flash
- Gemini 3.6 Flash
- Gemini 3.5 Flash
- Gemini 3.5 Transcribe Live
- Gemini 3.5 Transcribe
- Gemini 3.5 Flash-Lite
- Gemini Omni Flash
- Gemini Omni Flash Preview
- Gemini 3.8 Live, Gemini 3.8 Live Extended Thinking, and Gemini 3.1 Flash Live Preview
- Gemini 3.5 Live Translate
- Gemini 3.1 Flash-Lite
- Gemini 3.1 Pro Preview
- Gemini 3.1 Flash Image (Nano Banana 2) 🌟
- Gemini 3.1 Flash Lite Image (Nano Banana 2 Lite) 🌟
- Gemini 3.1 Flash TTS Preview
- Gemini 3 Flash Preview
- Gemini 3 Pro Image (Nano Banana Pro) 🌟
- Gemini 2.5 Pro
- Gemini 2.5 Flash
- Gemini 2.5 Flash-Lite
- Gemini 2.5 Flash Native
- Audio Live API

Latest Gemini Model 2026

Link Access: <https://aistudio.google.com/docs/models>

The screenshot shows the Google AI Studio interface. The left sidebar contains navigation links: Google AI Studio, Documentation, API reference, Community, GET STARTED (Overview, Get started, API keys, Pricing, Coding agent setup), and MODELS (All models, Latest Gemini models). The main content area is titled 'Models' and 'Gemini 3'. It features six model cards arranged in a 2x3 grid:

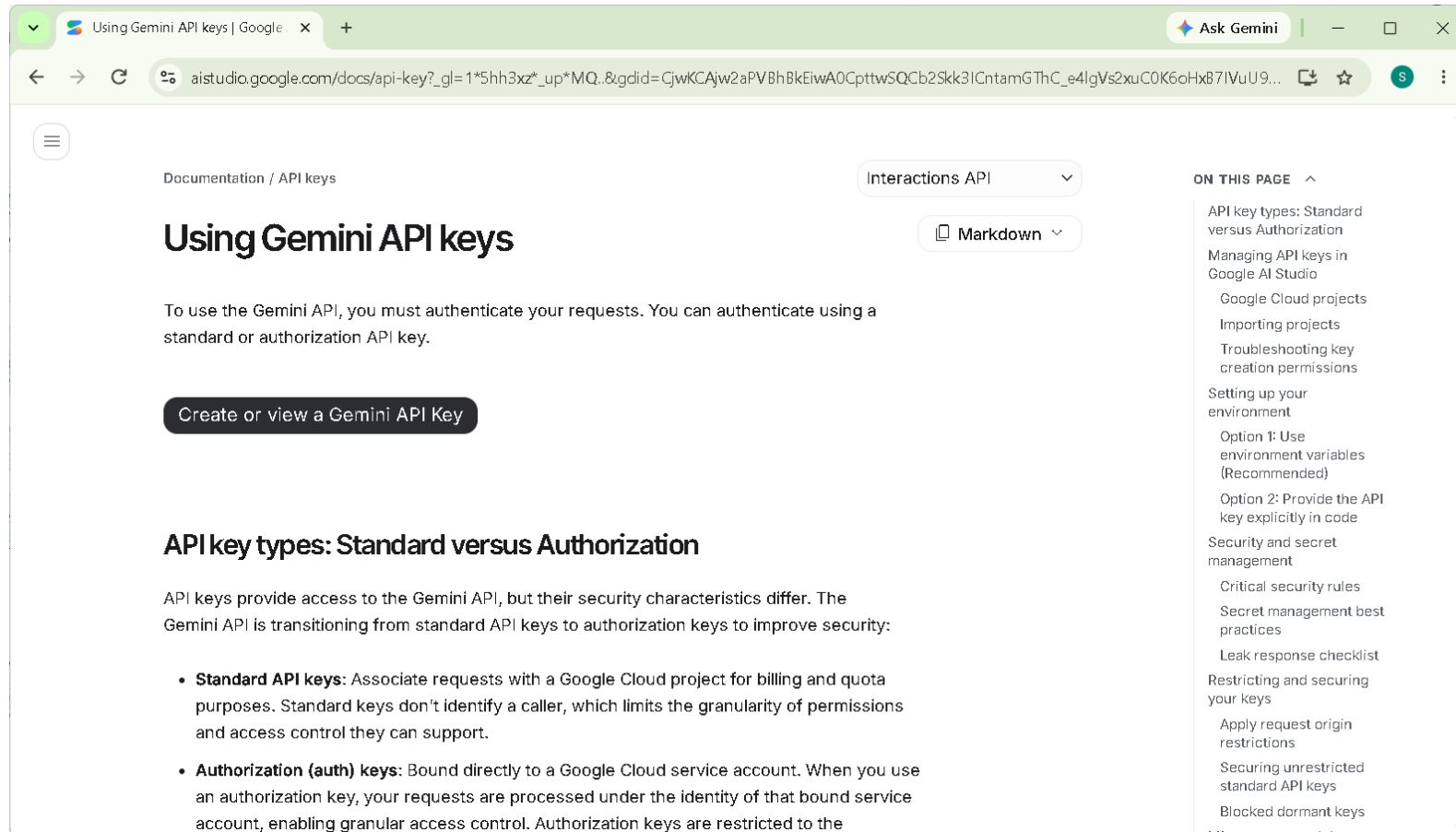
- Gemini 3.8 Flash**: Our most intelligent Flash model, engineered for long-horizon software engineering, autonomous agents, and complex enterprise workflows.
- Gemini 3.8 Live**: Default Live API model for most low-latency voice agent experiences without reasoning delays.
- Gemini 3.8 Live Extended Thinking**: High-reasoning Live API model for voice interactions, recommended when higher background reasoning is required.
- Gemini 3.7 Flash**: Our previous-generation Flash model for complex coding, agentic workflows, and reliable multi-step execution.
- Gemini 3.6 Flash**: Our previous-generation Flash model, balancing speed and multimodal capabilities across general agentic and...
- Gemini 3.5 Flash**: Our legacy Flash model, providing baseline speed and foundational performance for routine, high-throughput workloads.

On the right side of the main content area, there is a 'Markdown' button and a 'ON THIS PAGE' section with a list of links: Gemini 3, All Gemini 3 models, Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Gemini 2.5 Pro, Audio models, Generative media models, Music generation models, Tool and agent models, Specialized task models, Previous models, Model version name patterns, and Model deprecations.

The browser's address bar shows the URL: https://aistudio.google.com/docs/models/gemini-3.8-live?_gl=1*5hh3xz*_up*MQ..&gclid=CjwKCAjw2aPV8hBkEiwA0CpttwSQCb25kk3lCntamGThC_e4lgVs2xuC0K6oHx87lVU90mG1z1gBoC2yEQAvD_BwE&gbrad=0AAAAACn9t647...

Using Gemini API keys

Link Access: <https://aistudio.google.com/docs/api-key>



The screenshot shows a web browser window with the URL https://aistudio.google.com/docs/api-key?_gl=1*5hh3xz*_up*MQ..&gclid=CjwKCAjw2aPVBhBkEiwA0CpttwSQCb2Skk3ICntamGThC_e4lgVs2xuCOK6oHxB7IVuU9.... The page title is "Using Gemini API keys" under the "Documentation / API keys" section. The main heading is "Using Gemini API keys". Below it, a paragraph states: "To use the Gemini API, you must authenticate your requests. You can authenticate using a standard or authorization API key." A button labeled "Create or view a Gemini API Key" is visible. The section "API key types: Standard versus Authorization" follows, with a paragraph explaining that API keys provide access to the Gemini API but have different security characteristics, and that the Gemini API is transitioning from standard API keys to authorization keys. A bulleted list details the two types:

- Standard API keys:** Associate requests with a Google Cloud project for billing and quota purposes. Standard keys don't identify a caller, which limits the granularity of permissions and access control they can support.
- Authorization (auth) keys:** Bound directly to a Google Cloud service account. When you use an authorization key, your requests are processed under the identity of that bound service account, enabling granular access control. Authorization keys are restricted to the

 On the right side, there is a "ON THIS PAGE" table of contents with links to various sections: "API key types: Standard versus Authorization", "Managing API keys in Google AI Studio", "Google Cloud projects", "Importing projects", "Troubleshooting key creation permissions", "Setting up your environment", "Option 1: Use environment variables (Recommended)", "Option 2: Provide the API key explicitly in code", "Security and secret management", "Critical security rules", "Secret management best practices", "Leak response checklist", "Restricting and securing your keys", "Apply request origin restrictions", "Securing unrestricted standard API keys", "Blocked dormant keys", and "Migrate to authorization keys".

Create Gemini API Keys 2026

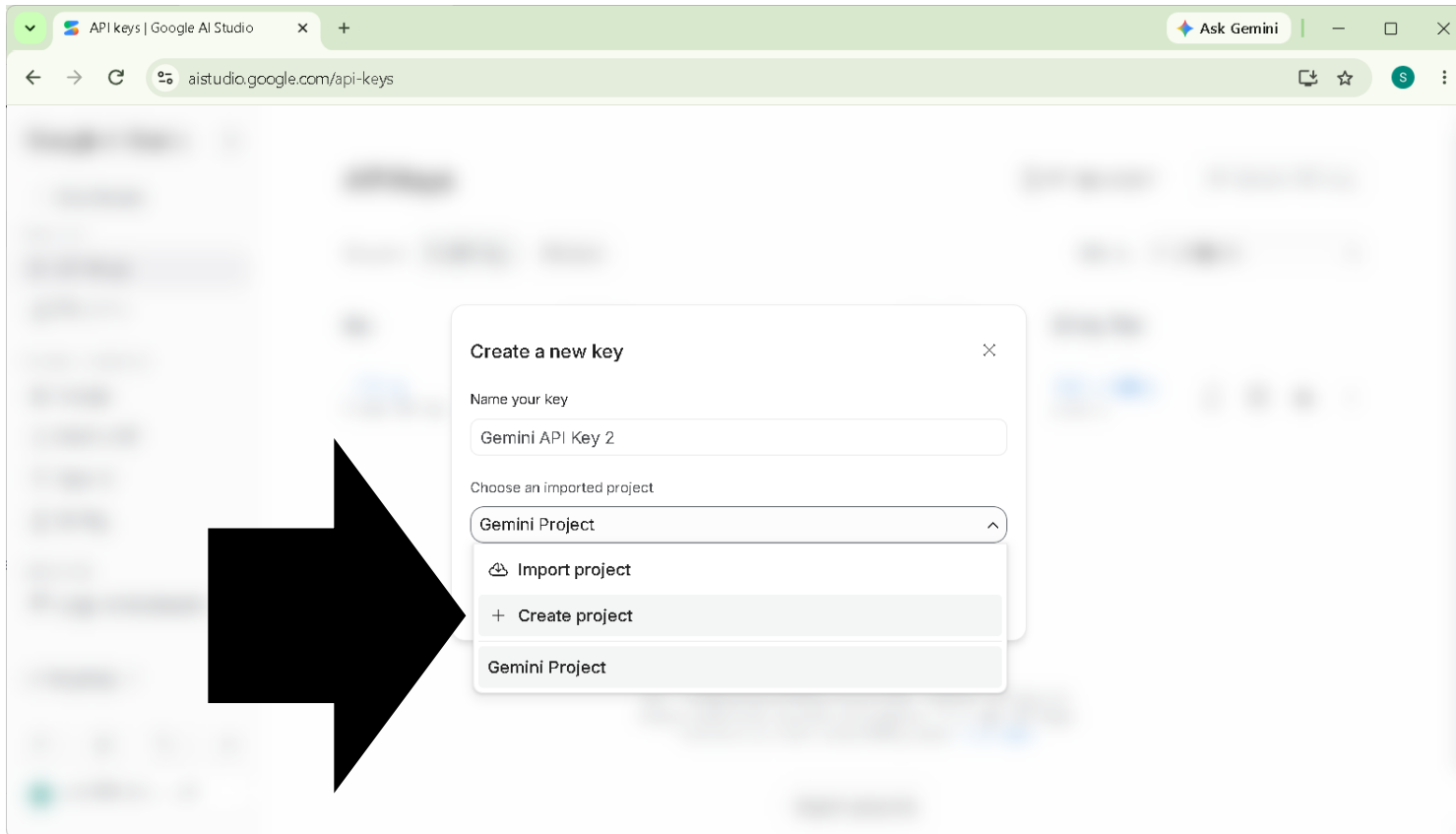
Link Access: <https://aistudio.google.com/docs/api-key>

The screenshot displays the Google AI Studio interface for managing API keys. The left sidebar contains navigation links for Dashboard, PROJECT (API Keys, Projects), USAGE & BILLING (Usage, Rate Limit, Spend, Billing), and OBSERVE (Logs and Datasets). The main content area is titled 'API Keys' and includes a 'Create API key' button. Below the title, there are filters for 'Group by' (API Key, Project) and 'Filter by' (All projects). A table lists the API keys with columns for Key, Project, Created, and Billing Tier. One key is listed: '...CJHg' (Gemini API Key) associated with 'Gemini Project' (gen-lang-client-0201611053), created on 'Sep 16, 2026', with a 'Free tier' billing tier. Below the table, a message states: 'Can't find your API keys here? This list only shows API keys for projects imported into Google AI Studio. Import other projects to manage their associated API Keys. You can also create a new API Key above. [Learn more](#)'. An 'Import projects' button is located at the bottom right.

Key	Project	Created	Billing Tier
...CJHg Gemini API Key	Gemini Project gen-lang-client-0201611053	Sep 16, 2026	Free tier

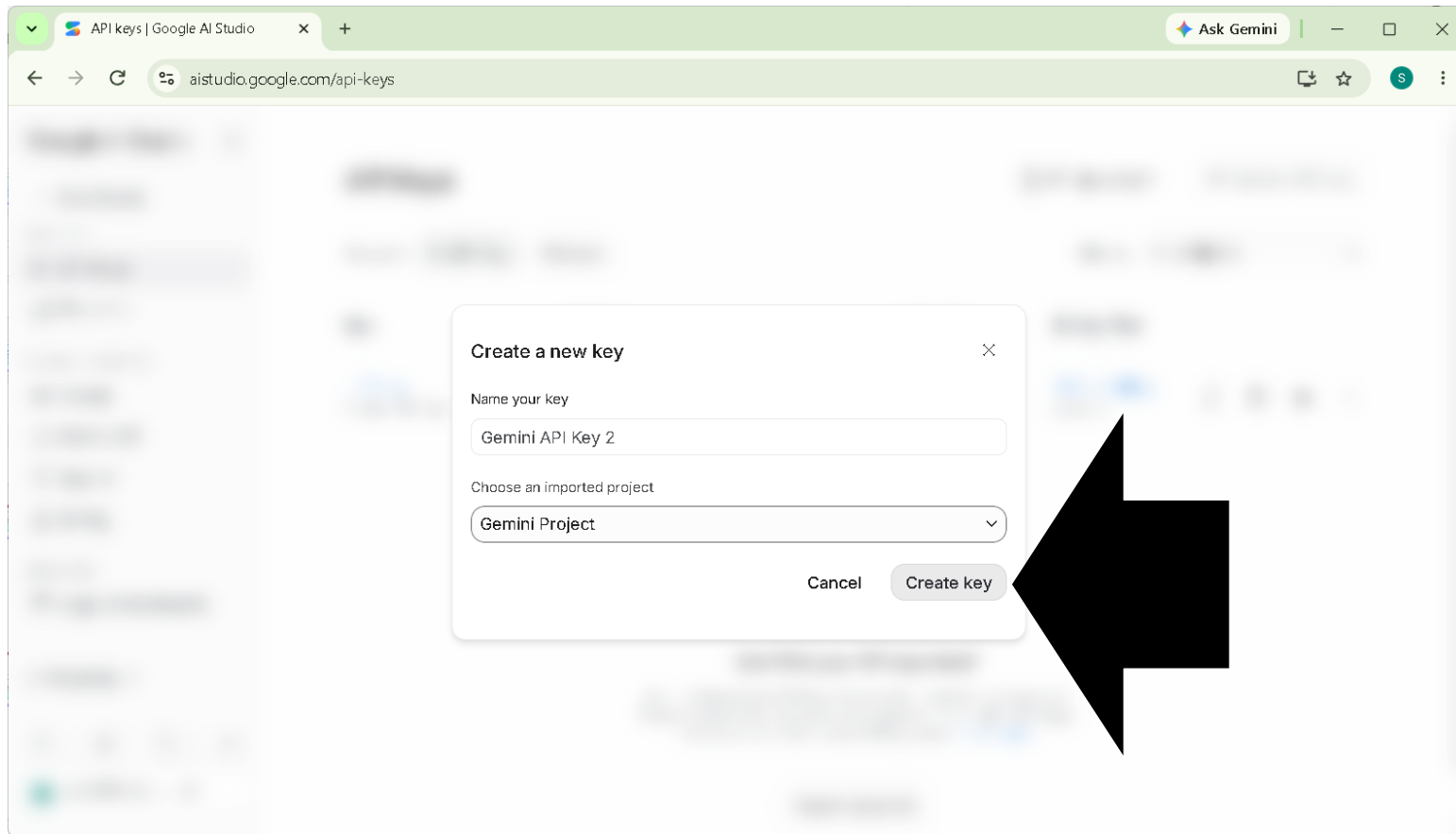
Create Google Project 2026

Link Access: <https://aistudio.google.com/docs/api-key>



Create Gemini API Keys 2026

Link Access: <https://aistudio.google.com/docs/pricing>



Copy Gemini API Keys 2026

Link Access: <https://aistudio.google.com/docs/pricing>

API key details

✕

ⓘ

For details on using auth keys with Gemini API, see our [API key documentation](#).

API Key

Ab8RN6lrFWP1-s-7niQi-YQ8g...

📋

Name

Gemini API Key

📋

Project name

projects/1060792062953

📋

Project number

1060792062953

📋

Delete key

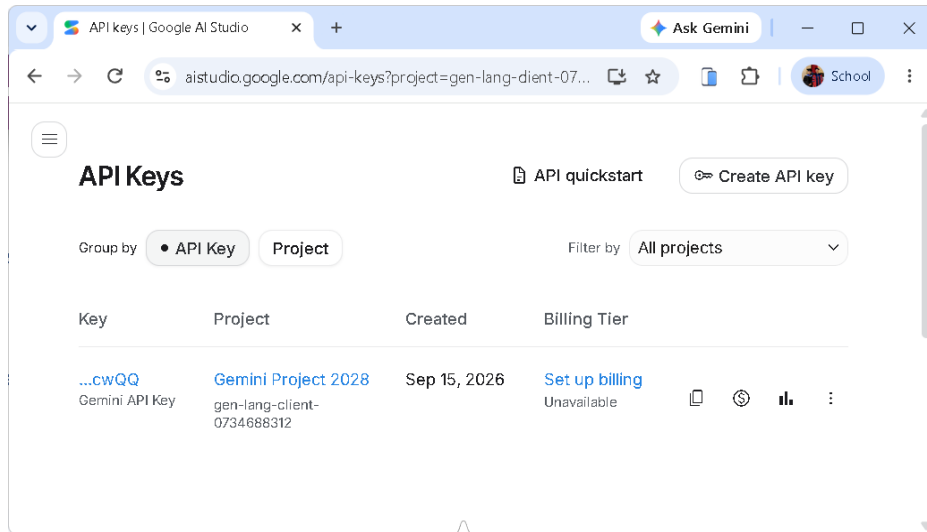
Copy cURL quickstart

Copy key

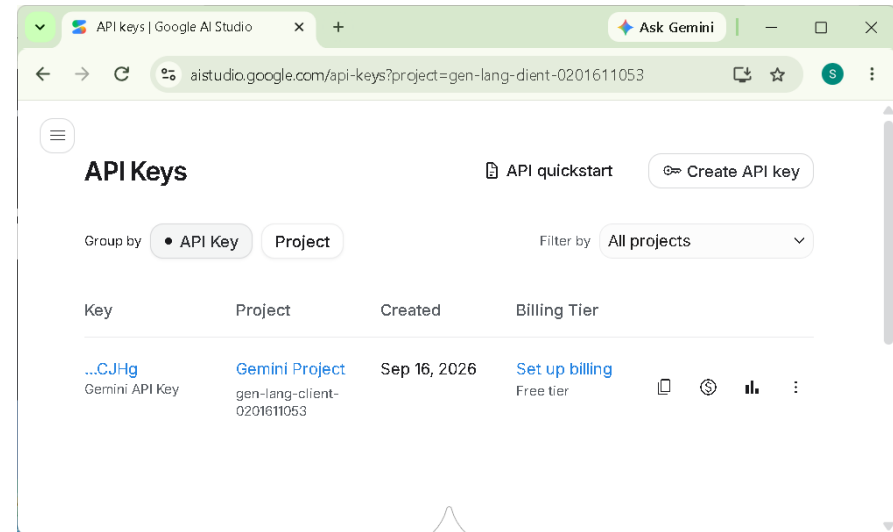
Gemini API Key 2026

Perhatikan pada status **Billing Tier**, jika menggunakan Account Gmail Unklab maka akan muncul status **Unavailable** (*Tidak bisa digunakan*). Mohon bisa gunakan email account pribadi.

URL: <https://aistudio.google.com/api-keys>



**Menggunakan Account
email UNKLAB**

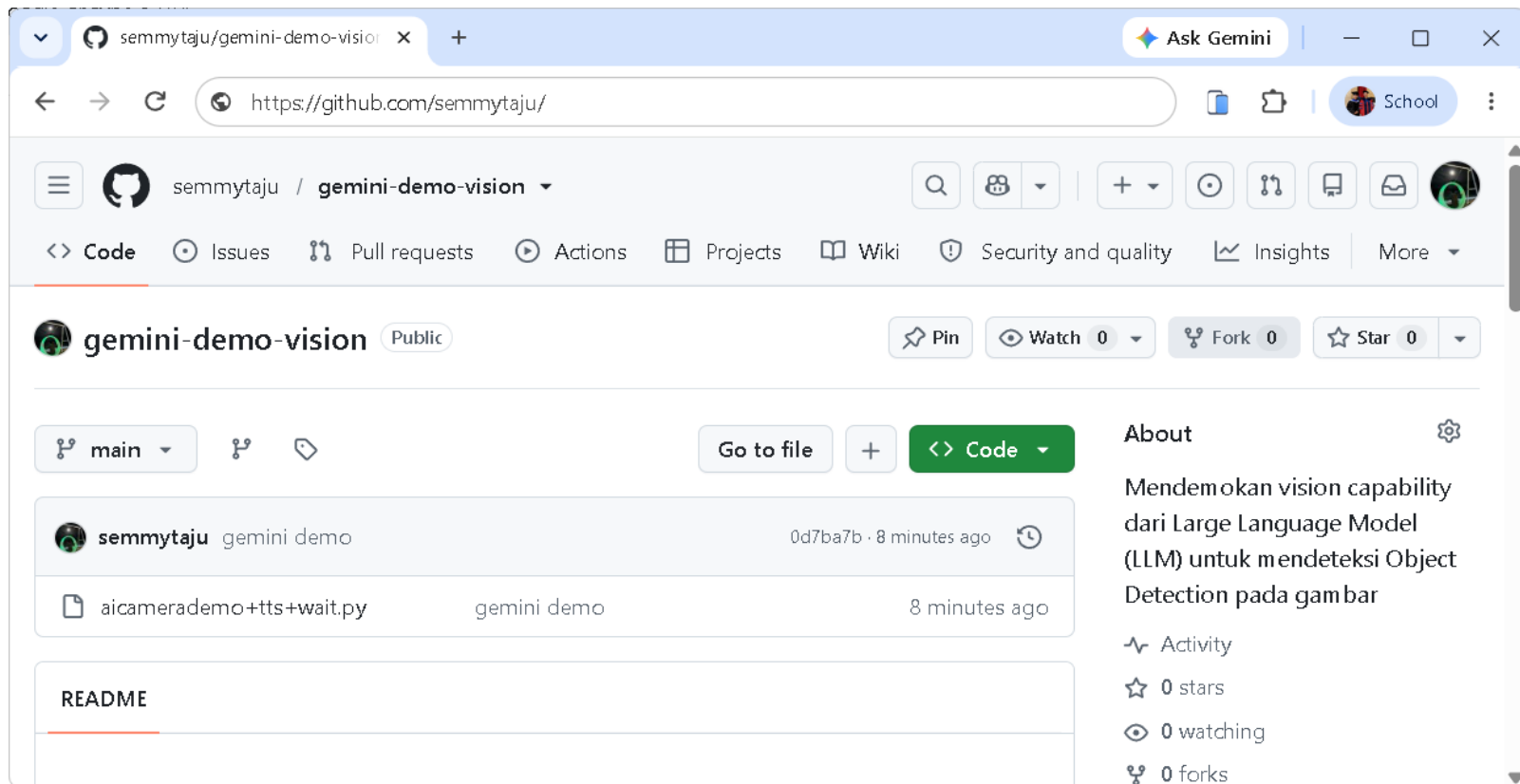


**Menggunakan Account
email Pribadi**

Download Code di Github

Link Github: <https://github.com/semmytaju>

Cari dan clone repository “**gemini-demo-vision**”.



Python Package Dependencies

Python Package Dependencies :

- Python: 3.9.7
- SpeechRecognition: 3.10.0
- gTTS: 2.5.2
- PyGame: 2.3.0
- NumPy: 1.23.3

Buka CMD, Run Administrator & Install:

- pip install SpeechRecognition
- pip install gtts
- pip install pygame

Gemini 3.8 Flash (2026)

Clone using the web URL:

<https://github.com/semmytaju/gemini-demo-vision.git>

Compile code:

>python "aicamerademo+tts+wait.py"

```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER\Downloads\gemini>python "aicamerademo+tts+wait.py"
pygame 2.3.0 (SDL 2.24.2, Python 3.9.7)
Hello from the pygame community. https://www.pygame.org/contribute.html
[
C:\Users\USER\Downloads\gemini>
```

Update Your API Key:

```
9  import tempfile
10 import time
11 import pygame
12 from gtts import gTTS
13
14 # Gemini API Key
15 API_KEY = "INPUT-YOUR-API-KEY"
16
17 # Gemini 3.5 Flash
18 #GEMINI_URL = f"https://generativelan
19
```

Gemini 3.8 Flash (2026)

AI Camera Analyzer



• Camera Active

Start Camera Analyze

Instruction:

Apa yang kamu lihat???

Response:

Sebuah tangan sedang memegang **mouse komputer berwarna biru**.

Interval between 2 requests: 5 Start Auto

Code di atas bisa di modifikasi, sesuai dengan project yang akan kalian buat. Topik dari project ini yaitu **Pengenalan Object di dalam bangunan.**

Demo

CLASS ACTIVITY #2

(Machine Learning (ML) and Large Language Model (LLM))

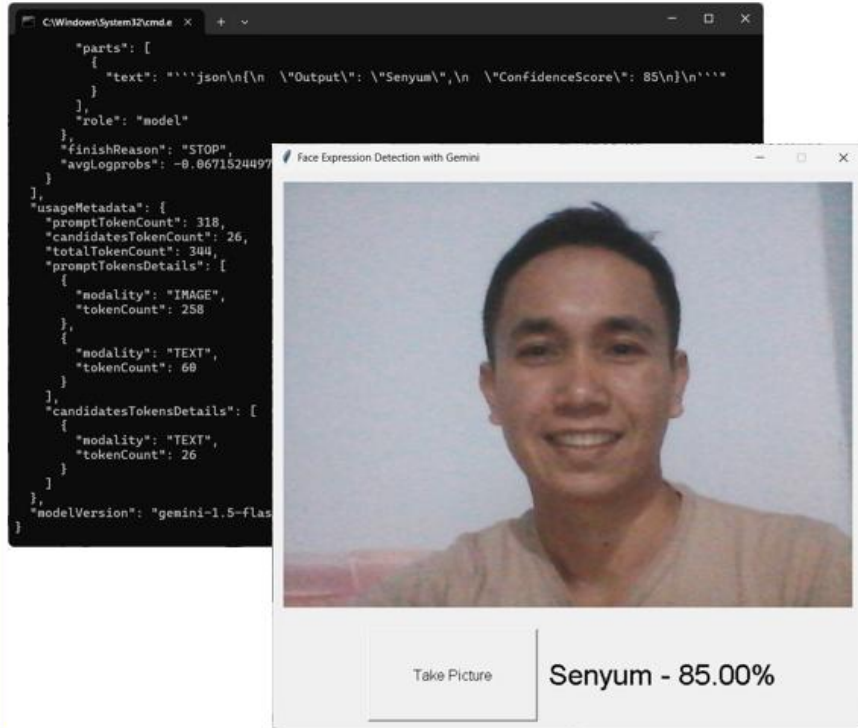
Comparing Machine Learning (ML) and Large Language Model (LLM) for Object Detection

Perbandingan Machine Learning (ML) vs Large Language Model (LLM) dalam Object Detection

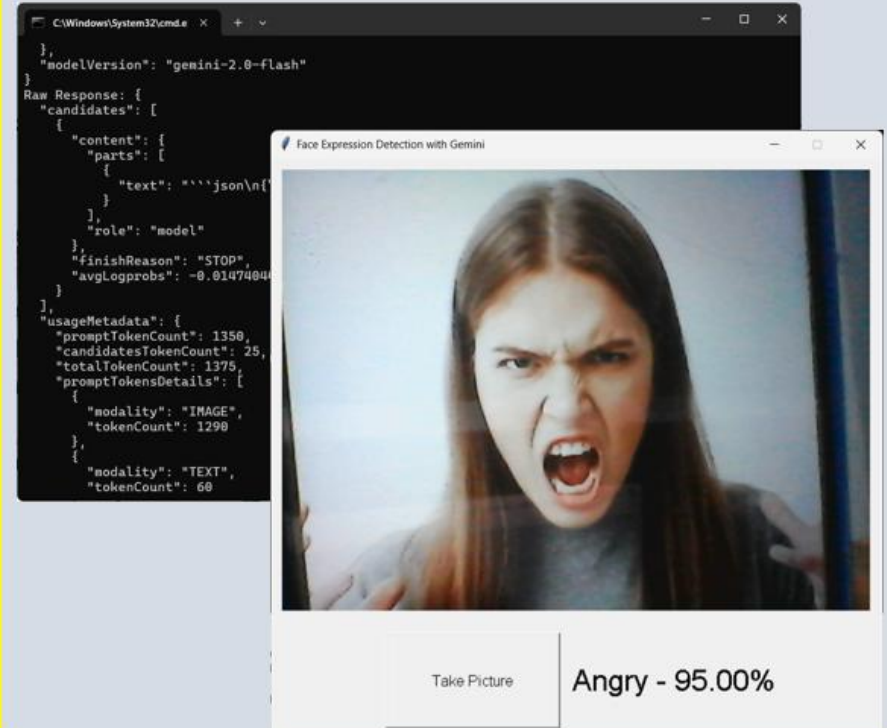
2025

ML Traditional vs LLM using Gemini

Gemini 1.5 Flash



Gemini 2.0 Flash



Face Expression Using Gemini 2.0 Flash

Menurut kalian model mana yang lebih baik dalam membedakan ekspresi wajah manusia???

Learning Objectives

After completing this activity, students are expected to be able to:

- 1) Menjelaskan konsep dasar **Object Detection**.
- 2) Membedakan pendekatan **Machine Learning (ML) Object Detection** dengan **Large Language Model (LLM) berbasis Vision**.
- 3) Membandingkan input, proses, output, kelebihan, dan keterbatasan kedua pendekatan.
- 4) Menganalisis hasil yang dihasilkan oleh ML dan LLM dari gambar yang sama.
- 5) Menentukan pendekatan yang sesuai berdasarkan kebutuhan suatu aplikasi AI.

Activity Description

In this activity, students will investigate how **Machine Learning (ML)** and **Large Language Models (LLMs)** can be used to identify and understand objects in an image. Students will use the same image as input and compare the results obtained from:

ML Object Detection

Menggunakan model Computer Vision khusus untuk mendeteksi dan menentukan lokasi objek dalam gambar, misalnya model berbasis YOLO.

LLM Vision / Visual Understanding

Menggunakan LLM yang memiliki kemampuan memahami gambar untuk mengidentifikasi objek dan menjelaskan isi gambar menggunakan natural language.

Fokus utama aktivitas:

Apa perbedaan antara "*mendeteksi objek*" dan "*memahami isi gambar*"? Dimana tujuan aktivitas bukan sekadar menentukan model mana yang **lebih baik**, tetapi memahami kapan dan mengapa masing-masing pendekatan lebih sesuai digunakan.

Comparison Analysis

Bandingkan hasil kedua pendekatan berdasarkan beberapa aspek berikut.

1. Kemampuan Mendeteksi Objek

- ❖ Berapa banyak objek yang berhasil dideteksi ML?
- ❖ Berapa banyak objek yang berhasil diidentifikasi LLM?
- ❖ Apakah ML dan LLM menemukan objek yang sama?
- ❖ Objek apa yang hanya ditemukan oleh ML?
- ❖ Objek apa yang hanya ditemukan oleh LLM?

1. Kemampuan Memahami Konteks

- ❖ Apakah ML dapat menjelaskan hubungan antarobjek?
- ❖ Apakah LLM dapat menjelaskan apa yang sedang terjadi dalam gambar?
- ❖ Apakah LLM dapat memberikan penjelasan yang lebih kontekstual?

1. Akurasi dan Kesalahan

- ❖ Objek yang salah dideteksi oleh ML.
- ❖ Objek yang terlewat oleh ML.
- ❖ Objek yang salah diidentifikasi oleh LLM.
- ❖ Kemungkinan hallucination dari LLM.

1. Kecepatan dan Kebutuhan Komputasi

- ❖ Kecepatan pemrosesan
- ❖ Kebutuhan hardware
- ❖ Kebutuhan koneksi internet/cloud
- ❖ Kebutuhan sumber daya komputasi
- ❖ Kebutuhan pecahayaan gambar
- ❖ Kebutuhan jarak/resolusi gambar

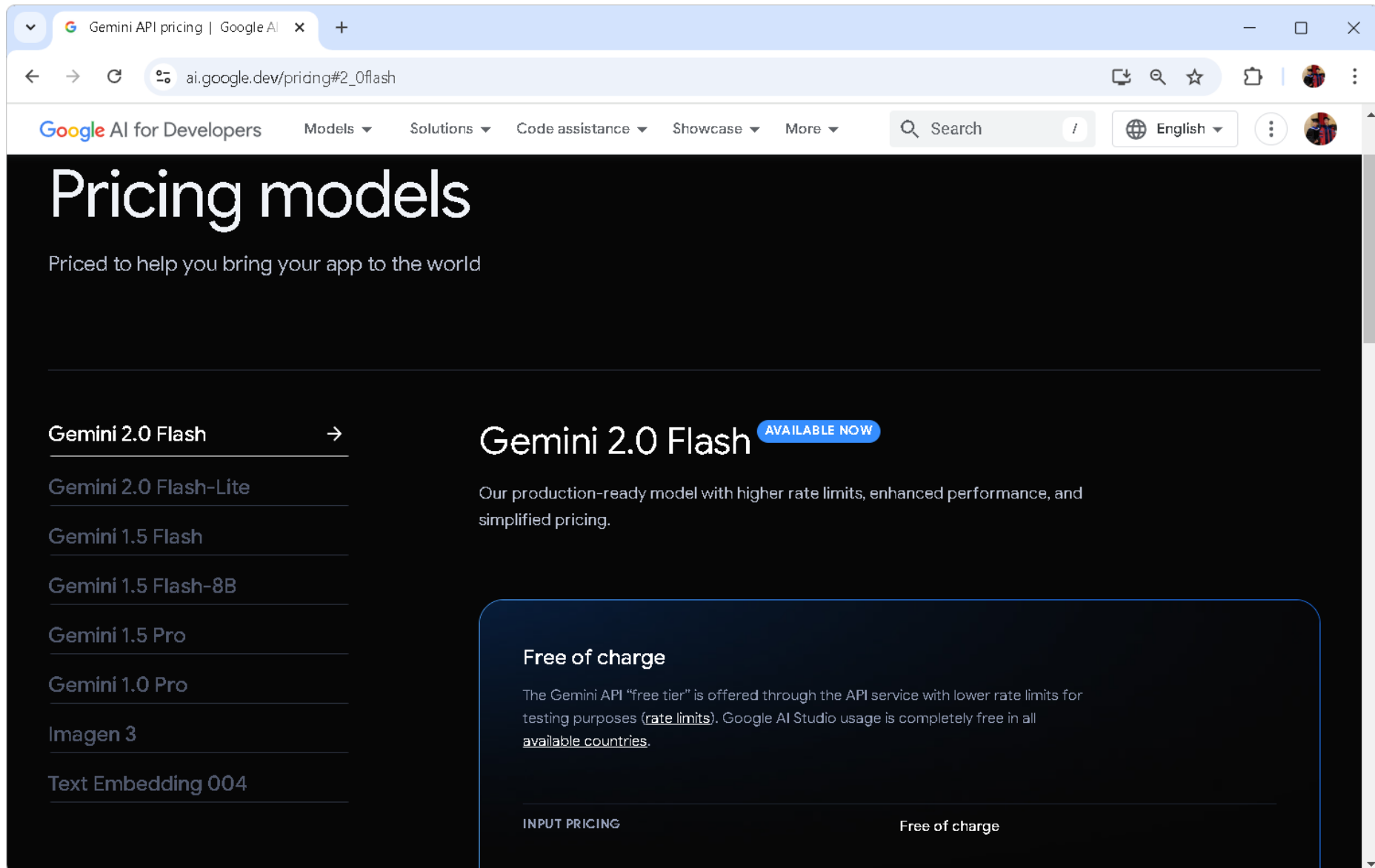
Critical Thinking Questions

- 1) Mengapa model ML khusus Object Detection biasanya lebih sesuai untuk masalah dunia nyata untuk deteksi **objek secara presisi**?
- 2) Mengapa LLM dapat memberikan penjelasan yang lebih panjang dan kontekstual mengenai sebuah gambar?
- 3) Apakah LLM dapat sepenuhnya menggantikan model Object Detection khusus? Jelaskan alasan kalian.
- 4) Jika kalian diminta membuat **sistem kamera keamanan berbasis AI**. Apakah kalian akan memilih: ML Object Detection?, LLM? Kombinasi ML + LLM? Jelaskan pilihan yang menurut anda lebih sesuai.
- 5) Jika kalian ingin membuat **AI Assistant** yang dapat menerima foto dari users dan menjawab pertanyaan mengenai foto tersebut. Pendekatan mana yang lebih sesuai? Jelaskan.
- 6) Menurut anda, apa perbedaan paling mendasar antara: “**Mendeteksi sebuah objek**” dan “**Memahami sebuah gambar**”?

Exercise *for* Students

(Memanfaatkan OpenAI GPT-4o dan/atau Gemini 1.5 Flash untuk menganalisis ekspresi wajah students dan untuk mengetahui tingkat kepuasan di dalam kelas)

Google Gemini 2.0 Flash (New Model)



The screenshot shows a web browser window with the URL `ai.google.dev/pricing#2_0flash`. The page is titled "Pricing models" and features a sidebar with links to various models: Gemini 2.0 Flash, Gemini 2.0 Flash-Lite, Gemini 1.5 Flash, Gemini 1.5 Flash-8B, Gemini 1.5 Pro, Gemini 1.0 Pro, Imagen 3, and Text Embedding 004. The main content area highlights the Gemini 2.0 Flash model with an "AVAILABLE NOW" badge. It describes the model as production-ready with higher rate limits and simplified pricing. A "Free of charge" section explains that a free tier is available through the API service with lower rate limits for testing purposes, and that Google AI Studio usage is completely free in all available countries. At the bottom, a table shows the input pricing for the Gemini 2.0 Flash model as "Free of charge".

Gemini API pricing | Google AI

ai.google.dev/pricing#2_0flash

Google AI for Developers Models Solutions Code assistance Showcase More Search English

Pricing models

Priced to help you bring your app to the world

- Gemini 2.0 Flash →
- Gemini 2.0 Flash-Lite
- Gemini 1.5 Flash
- Gemini 1.5 Flash-8B
- Gemini 1.5 Pro
- Gemini 1.0 Pro
- Imagen 3
- Text Embedding 004

Gemini 2.0 Flash AVAILABLE NOW

Our production-ready model with higher rate limits, enhanced performance, and simplified pricing.

Free of charge

The Gemini API “free tier” is offered through the API service with lower rate limits for testing purposes ([rate limits](#)). Google AI Studio usage is completely free in all [available countries](#).

INPUT PRICING	Free of charge
---------------	----------------

Gemini API Pricing

Gemini API pricing | Google AI for Developers

Link: <https://ai.google.dev/pricing>

Saya hanya menggunakan versi free dari Gemini 1.5 Flash API

Google AI for Developers

Models

Solutions

Code assistance

Showcase

Community

Search

Gemini API docs

API Reference

SDKs

Pricing

Cookbook

Pricing models

Priced to help you bring your app to the world

Gemini 1.5 Flash

Gemini 1.5 Flash-8B

Gemini 1.5 Pro

Gemini 1.0 Pro

Text Embedding 004

Gemini 1.5 Flash

AVAILABLE NOW

Our fastest multimodal model with great performance for diverse, repetitive tasks and a 1 million context window. Now generally available for production use.

Free of charge

The Gemini API "free tier" is offered through the API service with lower rate limits for testing purposes. Google AI Studio usage is completely free in all [available countries](#).

RATE LIMITS	
	15 RPM (requests per minute)
	1 million TPM (tokens per minute)
	1,500 RPD (requests per day)

Get Gemini API key

Go to: <https://ai.google.dev/gemini-api/docs/api-key>

Click di button “**Get a Gemini API key in Google AI Studio**”.

The screenshot shows the Google AI for Developers website. The browser address bar displays the URL `ai.google.dev/gemini-api/docs/api-key`. The navigation bar includes links for Google AI for Developers, Models, Solutions, Code assistance, Showcase, and Community. A blue sidebar on the left contains links for Gemini API docs, API Reference, SDKs, Pricing, and Cookbook. The main content area features a blue banner for Gemini 2.0 Flash Experimental, a breadcrumb trail (Home > Gemini API > Models), and the title 'Get a Gemini API key'. Below the title, a paragraph explains that an API key is needed to use the Gemini API and can be created in Google AI Studio. A blue button labeled 'Get a Gemini API key in Google AI Studio' is highlighted with a red checkmark. The section 'Set up your API key' follows, with text explaining that for initial testing, a hard-coded API key is acceptable but not secure for production use.

Overview

Get started

- Quickstart
- API keys
- Libraries
- Release notes
- Developer forum

Models

- Gemini
- Gemini 2.0
- Experimental models

Capabilities

- Text generation
- Vision

Gemini 2.0 Flash Experimental is now available! [Learn more](#)

Home > Gemini API > Models

Get a Gemini API key

To use the Gemini API, you need an API key. You can create a key with a few clicks in Google AI Studio.

[Get a Gemini API key in Google AI Studio](#)

Set up your API key

For initial testing, you can hard code an API key, but this should only be temporary since it is not secure. The rest of the page discusses how to set up your API key locally as an environment variable with different operating systems.

Create Gemini API Key

Google AI Studio: <https://aistudio.google.com/app/apikey>

Click button “**Create API Key**” untuk membuat API key yang baru. Disini saya sudah membuat sebuah API key yang akan digunakan pada CURL.

Google AI Studio Get API key

Get API key

API keys

Quickly test the Gemini API

API quickstart guide

```
curl "https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?key=GEMINI_API_KEY" \
-H 'Content-Type: application/json' \
-X POST \
-d '{
  "contents": [{
    "parts": [{"text": "Explain how AI works"}]
  }]
}'
```

Use code with caution.

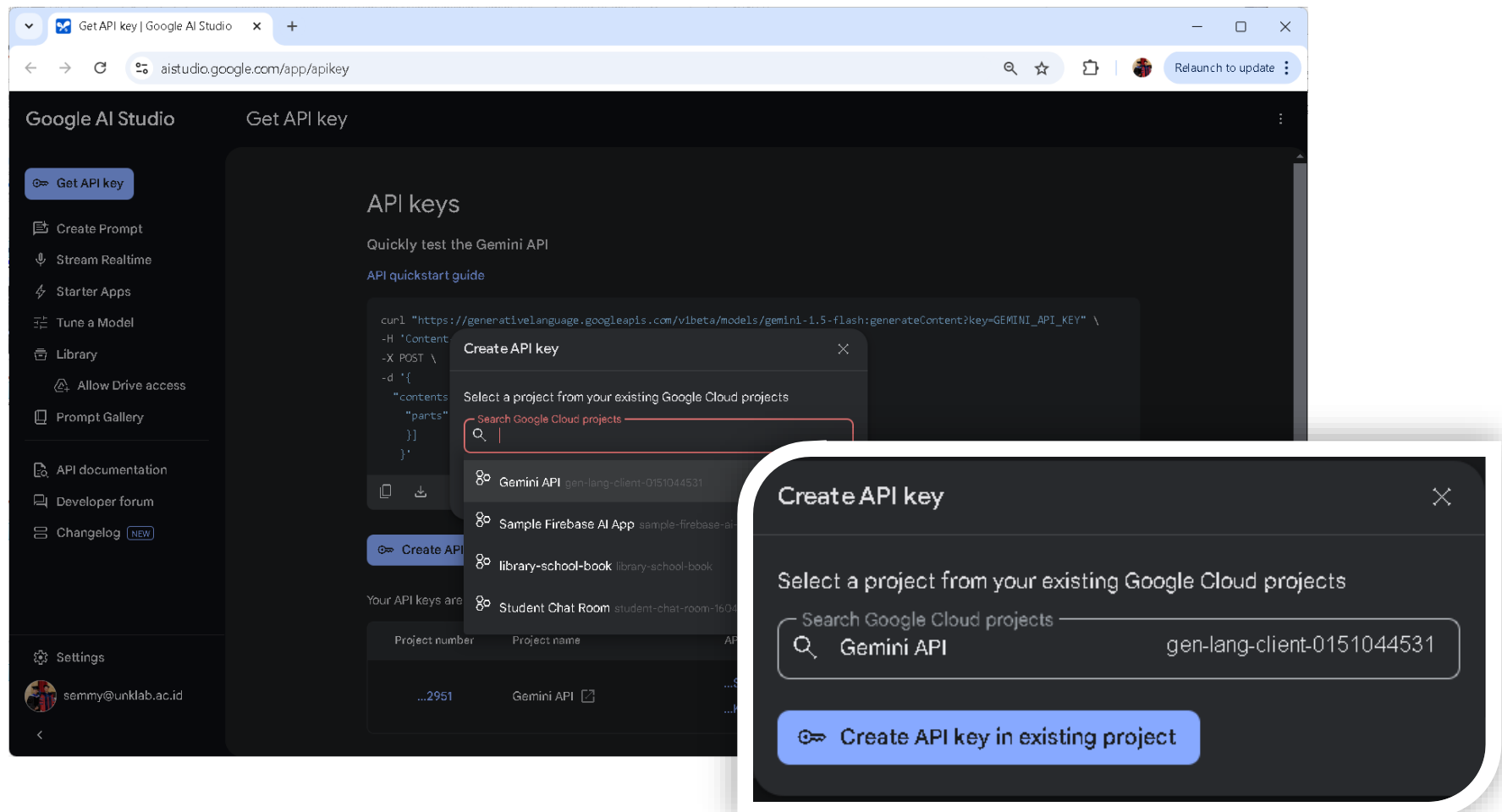
Create API key

Your API keys are listed below. You can also view and manage your project and API keys in Google Cloud.

Project number	Project name	API key	Created	Plan
...2951	Gemini API	...K9YY	Dec 29, 2024	Free of charge Set up Billing View usage data

Search Google Cloud projects

Select a project from your existing Google Cloud projects. Pilih Opsi “**Gemini API**”. Selanjutnya, click button “**Create API key in existing project**”.



The screenshot shows the Google AI Studio 'Get API key' page. The left sidebar contains navigation links: Get API key, Create Prompt, Stream Realtime, Starter Apps, Tune a Model, Library, Allow Drive access, Prompt Gallery, API documentation, Developer forum, and Changelog. The main content area is titled 'API keys' and includes a 'Quickly test the Gemini API' section with a cURL command and an 'API quickstart guide' link. A 'Create API key' modal is open, displaying a search bar for 'Search Google Cloud projects' and a list of projects. The 'Gemini API' project is selected, and the 'Create API key in existing project' button is highlighted.

API keys

Quickly test the Gemini API

API quickstart guide

```
curl "https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?key=GEMINI_API_KEY" \
-H "Content-Type: application/json" \
-X POST \
-d '{
  "contents": [
    {
      "parts": [
        {
          "text": "Hello, Gemini!"
        }
      ]
    }
  ]
}'
```

Create API key

Select a project from your existing Google Cloud projects

Search Google Cloud projects

Gemini API gen-lang-client-0151044531

Sample Firebase AI App sample-firebase-ai-app

library-school-book library-school-book

Student Chat Room student-chat-room-1604

Create API key

Select a project from your existing Google Cloud projects

Search Google Cloud projects

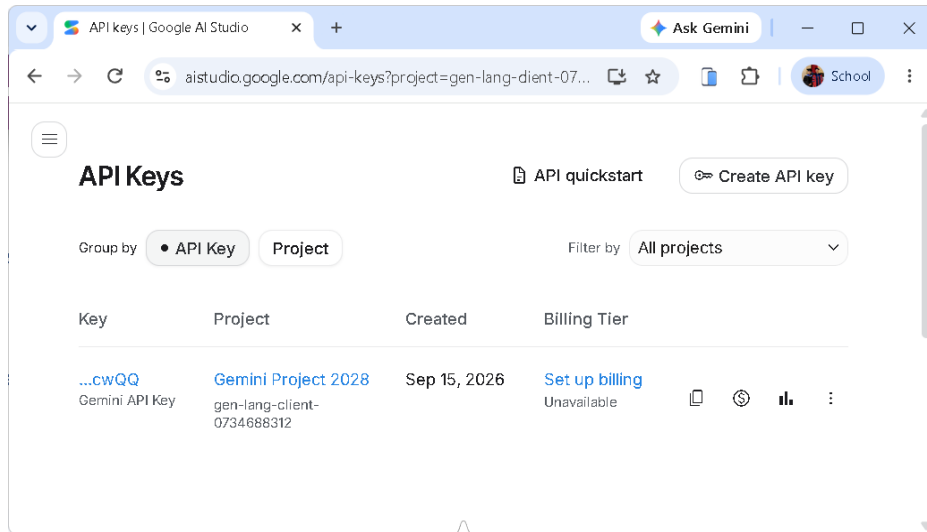
Gemini API gen-lang-client-0151044531

Create API key in existing project

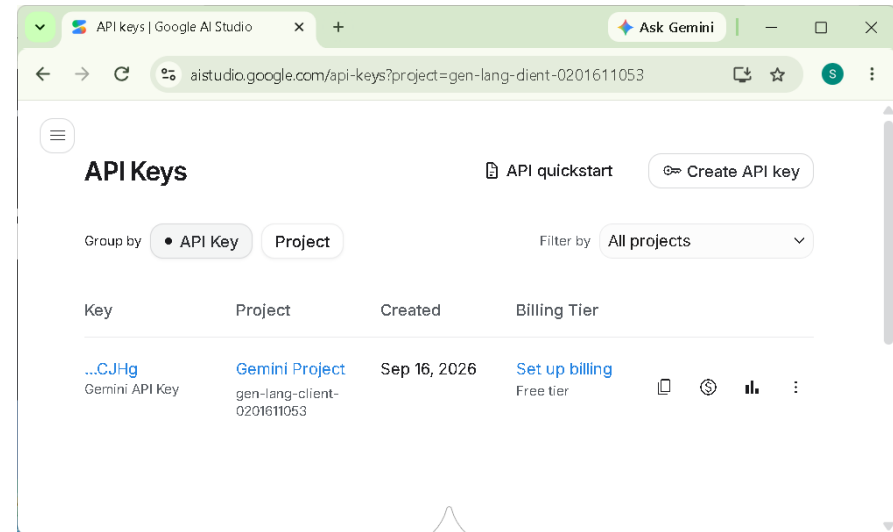
Gemini API Key

Perhatikan pada status **Billing Tier**, jika menggunakan Account Gmail Unklab maka akan muncul status **Unavailable** (*Tidak bisa digunakan*). Mohon bisa gunakan email account pribadi.

URL: <https://aistudio.google.com/api-keys>



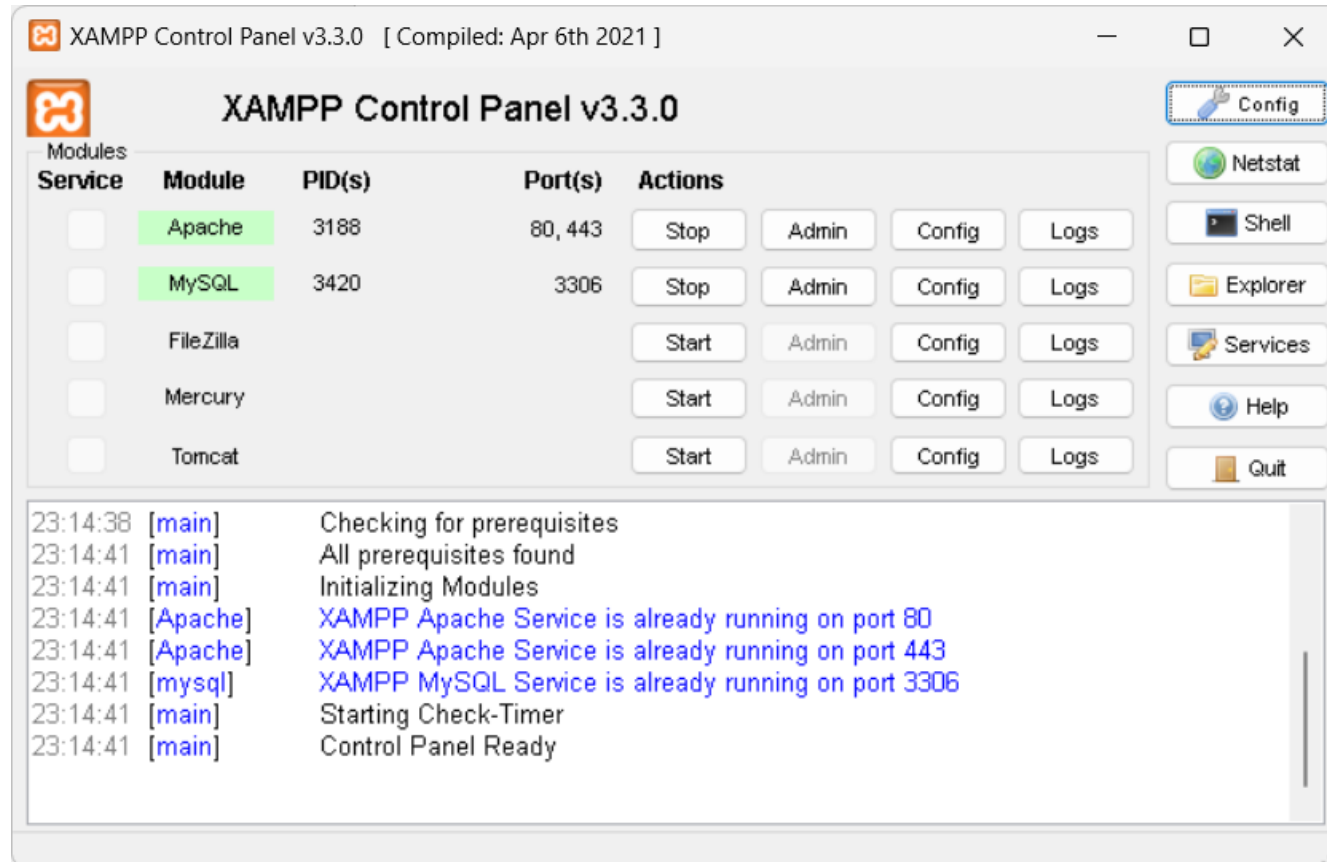
Menggunakan Account
email UNKLAB



Menggunakan Account
email Pribadi

XAMPP Control Panel

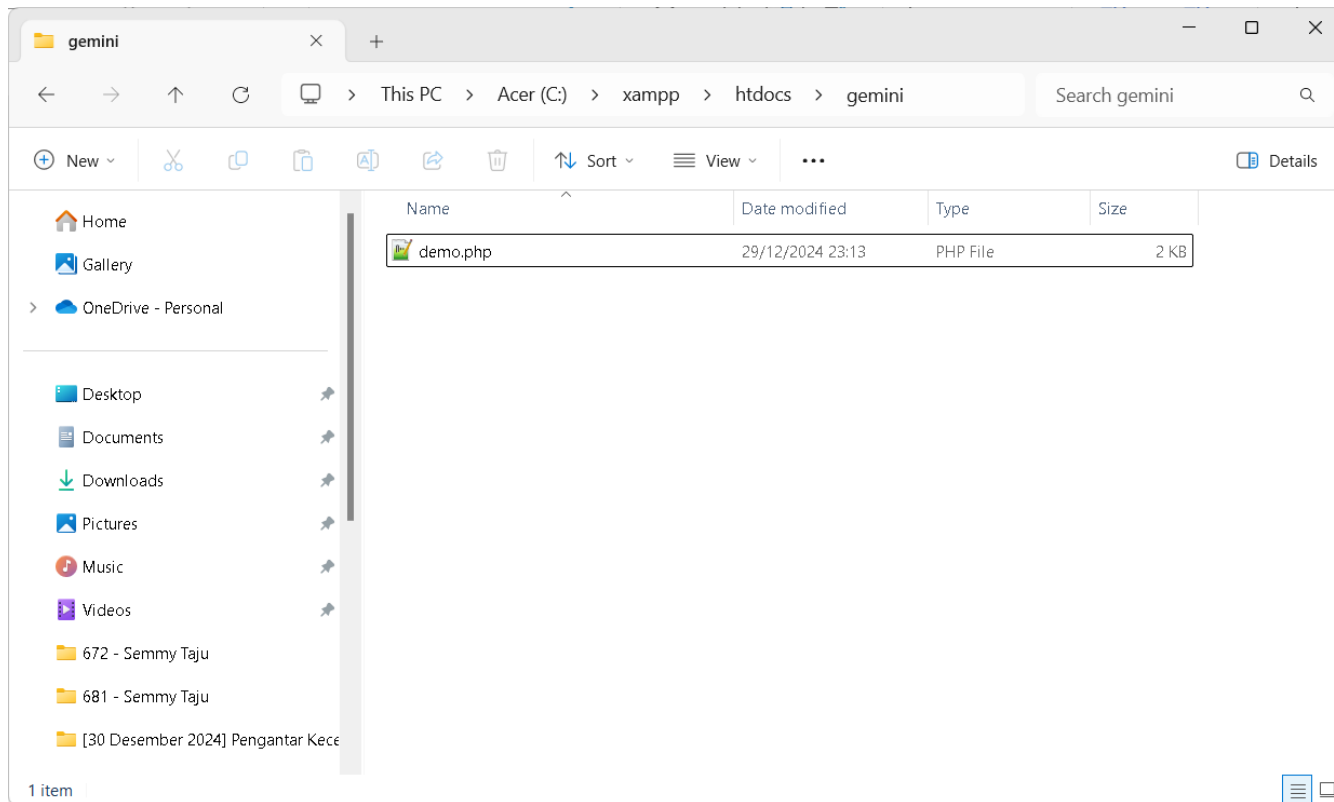
Lakukan instalasi xampp tool dan jalankan xampp control panel. Pastikan service untuk Apache sudah dijalankan.



Working Directory

Location: C:/xampp/htdocs/gemini/

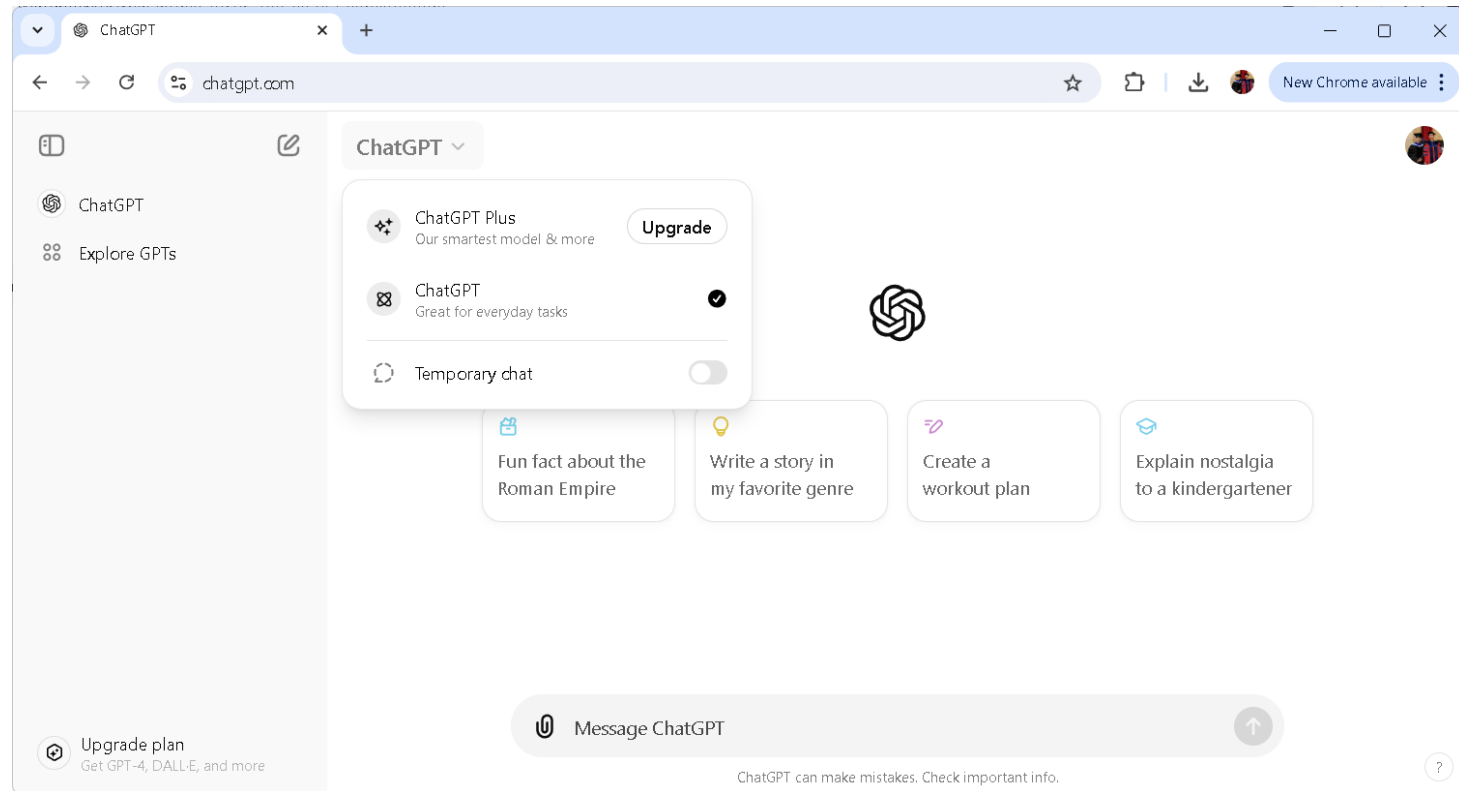
Buat sebuah folder “**gemini**” di htdocs directory dari XAMPP/Apache, terdapat sebuah file “**demo.php**” di dalam folder tersebut. Directory ini akan diakses dari web browser pada saat menjalankan project yang akan anda buat.





Lakukan pengambilan gambar di dalam class untuk semua students.

OpenAI GPT-4o (Vision Capability)



```

1 <?php
2
3 $apiKey = "AIzaSyDkZJtDRa76RlQEQvQs3nTAF0tUD2G4iVBHU"; // Gantilah dengan kunci API Google Anda
4 $imagePath = "C:/Users/USER/Downloads/monkey.jpg";
5
6 // Encode gambar ke base64
7 $imageData = base64_encode(file_get_contents($imagePath));
8
9 // URL endpoint API
10 $url = "https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?key={$apiKey}";
11
12 // Data JSON yang akan dikirim
13 $data = [
14     "contents" => [
15         [
16             "parts" => [
17                 [
18                     "text" => "Apakah semua ada gambar manusia? hewan apa saja yang ditampilkan pada gambar?",
19                     "inline_data" => [
20                         "mime_type" => "image/png",
21                         "data" => $imageData
22                     ]
23                 ]
24             ]
25         ]
26     ];
27
28 // Konversi data ke JSON
29 $jsonData = json_encode($data);
30
31 // Inisialisasi cURL
32 $ch = curl_init($url);
33
34 // Setel opsi cURL
35 curl_setopt($ch, CURLOPT_HTTPHEADER, [
36     'Content-Type: application/json',
37 ]);
38 curl_setopt($ch, CURLOPT_POST, true);
39 curl_setopt($ch, CURLOPT_POSTFIELDS, $jsonData);
40 curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
41
42 // Eksekusi permintaan cURL
43 $response = curl_exec($ch);
44
45 // Periksa apakah ada error
46 if (curl_errno($ch)) {
47     echo "Error: " . curl_error($ch);
48 } else {
49     echo $response;
50 }
51
52 // Tutup cURL
53 curl_close($ch);
54
55 >>

```

Google Gemini 1.5 Flash (Vision Capability)

**Buatlah versi website
kalian sendiri !!!**

Execute/Run di browser:

<http://localhost/gemini/demo.php>

Query Prompt VLM:

- 1. “Dari gambar yang diinputkan, berapa persentase students yang puas dan tidak puas terhadap materi kuliah hari ini?”**
- 2. Ada berapa banyak students di dalam kelas?**
- 3. Apakah Google Gemini bisa menganalisa expresi wajah setiap mahasiswa di dalam gambar?**
- 4. Buat perbandingan Google Gemini 1.5 Flash (Vision Capability) dan Google Gemini 2.0 Flash (Vision Capability), model yang yang bagus untuk menganalisa gambar?**
- 5. Hitung berapa banyak male dan female?**

Apakah semua prompt bisa di jawab oleh AI?

Presentasi Kelompok

Melalui aktivitas ini, mahasiswa diharapkan memahami bahwa ML Object Detection dan LLM Vision memiliki fungsi dan karakteristik yang berbeda.

Setiap kelompok akan mendapatkan waktu untuk mempresentasikan hasil aktivitas. Presentasi harus mencakup:

- 1) Gambar yang digunakan.
- 2) Hasil Object Detection menggunakan ML.
- 3) Hasil analisis menggunakan LLM.
- 4) Perbedaan hasil ML dan LLM.
- 5) Kelebihan dan keterbatasan masing-masing.
- 6) Rekomendasi penggunaan.
- 7) Kesimpulan.

Minimal satu anggota kelompok harus melakukan demonstrasi eksperimen.

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

